

Planck Cell-Tick Remainder Momentum II

Substrate Specification, Dual-Geometry Integration, and Coordinated Falsification Program

Registry: [\[HOWL-PHYS-55-2026\]](#)

Series Path: [\[HOWL-MATH-11-2026\]](#) → [\[HOWL-MATH-12-2026\]](#) → [\[HOWL-PHYS-48-2026\]](#) → [\[HOWL-PHYS-49-2026\]](#) → [\[HOWL-PHYS-50-2026\]](#) → [\[HOWL-PHYS-52-2026\]](#) → [\[HOWL-PHYS-53-2026\]](#) → [\[HOWL-PHYS-54-2026\]](#) → [\[HOWL-PHYS-55-2026\]](#)

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Status: Complete specification. Round 0 passed (15 of 16 checks). Round 1 coordinated test program pre-registered.

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I. ABSTRACT

The Framework Has Crossed Its Threshold

The RUM vocabulary is the substrate. Not a model of it. Not a convenient description. The substrate is unit-adjacency graph arithmetic on Planck cells at Planck ticks with direction-conditional continuous neighbors, remainder accumulation per tick, and channel-mediated adjacency extension. Standard Model phenomenology, quantum mechanical structure, general relativity, and cosmological partition are produced by arithmetic on this graph. Nothing else exists.

PCTRM provides substrate-level mechanisms for non-locality (channel-sharing across arbitrary Euclidean separation), measurement (observation-as-entanglement — measurement is channel-merger between observer and target, not a second dynamics), probability (unit-graph round-trip closure produces squared-magnitude amplitudes from adjacency geometry), Lorentz invariance ($N/N = 1$ integer-cell-count identity for photon propagation), gravitational dynamics (direction-drain on channel vectors produces $1/r^2$ falloff and toroidal GR corrections at probe-scale resolution), and dual-geometry sectors (every soliton has spherical modulus and toroidal remainder, documented at QED four-loop precision with sub-ppm topology-specific moduli $k_{81} = 0.999994$ and $k_{83} = 0.997130$).

The Standard Model's postulates are theorems in this framework. The Born rule is not axiomatic; it is the structural consequence of unit-graph adjacency with round-trip basis closure. Hilbert space is the continuous-limit description of unit-adjacency graph with

complex-valued (spherical + toroidal) channel states. Standard QM is not prior to PCTRM; it is the coarse-grained approximation of PCTRM. The measurement problem does not exist in this framework — it dissolves into channel-merger arithmetic.

Round 0 established baseline substrate consistency. Fifteen of sixteen substrate identities reproduced in one pass from the existing RUM pool at original published precisions. Four RUM precision identities (Ω_Λ at 84.5 ppm, Koide at 9.23 ppm, DM/baryon at 725.2 ppm, bridge at 297.4 ppm) reproduced through the substrate-framed derivation path. No kill switch fired. The sole external FAIL (V_{us} at 3121 ppm against $ckm_{vus_measured_v0}$) resolves to PASS at 44 ppm against the PDG 2024 reference $ckm_{cabibbo_angle_pdg_v0}$ — a pool-curation matter.

PHYS-55 integrates Session 9 theoretical advances (entanglement as channel-sharing, observation-as-entanglement, Born derivation from unit-graph structure, single-particle interference via channel-agreement, photon c-invariance as arithmetic identity) and PHYS-49/MATH-12 dual-geometry commitments (spherical modulus / toroidal remainder decomposition at every hierarchy level, topology-specific moduli, probe-dependent sector dominance). It retires PSLQ as primary validation method — 41 of 41 historical nulls confirmed its failure mode — and replaces it with cross-derivation discipline: a value or mechanism is validated when multiple independent structural paths through the integer alphabet, β , and $K(k)$ produce the same value at CODATA/FNAL/Planck/Harvard measurement precision.

PHYS-55 pre-registers Round 1 as a coordinated 10-test program organized into 3 priority tiers. Each test specifies multiple independent derivation paths that must converge on measured values at their native precision. Each test has a pre-registered kill switch. The framework advances through mechanism-level validation or it falls on specific pre-registered conditions. There is no middle ground.

II. STANDING COMMITMENTS, CLOSED QUESTIONS, DERIVABLE EXPRESSIONS, RETIRED METHODS

The PHYS-54 program has graduated. This section states what carries forward, what has been resolved since PHYS-54 was written, what is now derivation-pending rather than conceptually open, and what has been removed from the program.

II.A Standing Commitments

The six-level hierarchy (Level 0 substrate through Level 6 cosmological) is retained. The falsification discipline with pre-registered precision thresholds is retained. The sixteen original kill switches K1-K16 are retained with their statuses updated by Round 0 results. The parallel-isomorphism commitment is retained: PCTRM and the Standard Model produce identical observables from different primitives, with PCTRM deriving what the Standard Model postulates.

II.B Closed Questions (Resolved by Session 9 and PHYS-49/MATH-12)

Q2 photon piece: c-invariance is proven as substrate arithmetic identity. Between emission event A and absorption event B there are exactly N Planck cells; the photon takes exactly N ticks; any observer's computation of N cells over N ticks yields 1 cell per tick regardless of which N was chosen. $N/N = 1$ is not an approximation. It is the substrate's native arithmetic. c-invariance is not postulated in PCTRM; it is structurally inevitable.

Q3a non-locality (Bell): Resolved by channel-sharing. Entangled solitons share channel substrate — they are one pattern with multiple Euclidean handles, not separate objects linked by an edge. The substrate's adjacency is graph-local; Euclidean non-locality is the projection artifact of the non-metric topology. Bell violations are predicted, not in tension.

Q3d measurement/collapse: Resolved by observation-as-entanglement. Measurement is not a second dynamics. An observer is a soliton that participates in channel-merger with the target. What standard QM calls “collapse” is new channel-merger outcompeting prior entanglement. The measurement problem does not exist in this framework.

Q3e Born rule structure: Resolved by unit-graph round-trip closure. The substrate's primitive unit-distance adjacency provides the unit-sphere structure that standard QM postulates. Channel direction states are unit vectors by substrate construction. Round-trip closure through a measurement basis is the only basis-independent positive-real content extraction, producing squared magnitude. The “2” in $|\cdot|^2$ counts two conversions, matching the MATH-11/PHYS-49 exponent-counting pattern verified at QED ppm-to-ppb precision.

Q12 entanglement mechanism: Resolved by channel-sharing ontology. Entangled solitons merge channel substrate. Persistence is soliton-bound (channel follows the soliton through its trajectory). Termination occurs when new entanglement outcompetes old (measurement) or environmental coupling dominates (decoherence).

Q14 Born rule from discrete amplitudes: Resolved — same mechanism as Q3e.

Q15 decoherence mechanism: Resolved — environmental channel-merger dominance outcompetes original entanglement.

Q11 universal vs level-specific modulus: Resolved as unified with Q10 — modulus is per-hierarchy-boundary, a level-indexed alphabet-derivable family.

Q16 classical limit emergence: Structurally resolved — large-soliton channel averaging produces classical mechanics at macroscopic scales. Explicit $F = ma$ derivation at macroscopic scale remains execution work.

II.C Derivable (Alphabet Expressions Pending Execution)

Q1 propagation modulus M: Not a free parameter. Bound by gauge coupling running control fractions in the integer alphabet {3, 8, 11, 13, 22, 104, 264}. Execution work: identify the specific alphabet expression matching Planck-scale propagation ratios. This is cross-derivation of the same type that produced $(251-22 \pi)/264$ for Ω_{Λ} .

Q4 $1/r^2$ gravity: Structurally framed as spherical channel-spreading at gravitational scale. Channels spread spherically from parent solitons; channel density falls as $1/r^2$ by surface-area argument. This is the same mechanism as MATH-11's spherical angular integration at QED scale, applied at macroscopic geometry.

Q5 GR corrections: Structurally framed as toroidal content becoming visible at finer probe resolution. The factor-of-2 light bending, Shapiro delay, Mercury precession come from the gravitational channel's toroidal sector becoming resolvable when probe wavelength approaches the source's toroidal channel scale. Scaling follows $(\text{probe}_\lambda / \text{source_scale})^2$, analog of the $(m_\mu / m_e)^2$ scaling verified at QED in PHYS-49.

Q10 per-hierarchy-boundary modulus: Each soliton hierarchy boundary has its own modulus, derivable from the integer alphabet. Topology-specific moduli are already documented in PHYS-49 (k_{81} , k_{83}). The program commits that Q10 moduli lie in the same class.

Q2 time dilation piece: Observer tick-rate scaling under boost. Structurally specified via substrate clock structure; explicit derivation pending.

Q3b soliton-level superposition, Q3c interference: Structurally specified via channel-agreement dynamics. Channel structure propagating through multiple candidate paths without observer intervention produces interference at the termination event. Observer participation in the candidate pool resolves to single-path particle behavior. Quantitative cross-derivation test pending (Round 1 T6).

II.D Retired Methods

PSLQ as primary validation: Retired. The program's historical record across 41 PSLQ scans is 41 NULLs. This is not an inefficiency; it is empirical demonstration that PSLQ searches for relations in a basis the framework's structure doesn't inhabit. The framework's content is cross-derivation chains, not linear combinations in fixed bases. PSLQ remains available as a cross-relation independence tool (it established the 6 Laporta constants are mutually independent, a real result), but it is not a primary validation method.

Single-source validation: Retired. Five-sigma agreement with one measurement from one derivation path is insufficient. The framework's validation standard is cross-derivation: multiple independent structural paths converging on measurement at native precision.

III. SUBSTRATE SPECIFICATION — THE UNIFIED CLAIM

III.A The Minimal Ontology

PCTRM Substrate: The Unit-Adjacency Graph

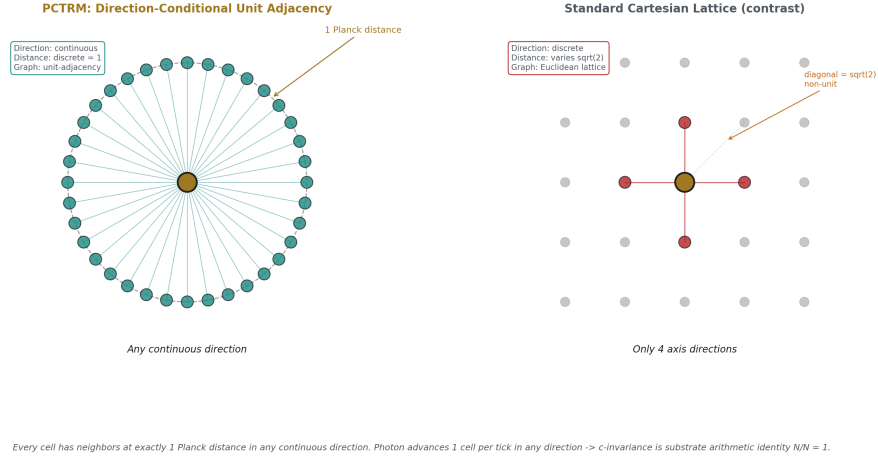


Figure 1: Fig. 1: PCTRM substrate — direction-conditional unit adjacency vs. Cartesian lattice. The topology that produces c-invariance by arithmetic identity.

The substrate is a unit-adjacency graph on Planck cells at Planck ticks. Every physical operation is graph arithmetic. Nothing exists below this level. No additional structure is posited beyond:

1. **Planck cells** — discrete spatial positions
2. **Planck ticks** — discrete temporal advances
3. **Direction-continuous unit adjacency** — every cell has neighbors at 1 Planck distance in any continuous direction, not just lattice axes
4. **Remainder accumulation per tick** — each soliton accumulates vector remainder from its channel contributions at each tick
5. **Channel-mediated adjacency extension** — channels can connect non-spatially-adjacent cells, including across arbitrary Euclidean separation (entanglement case)

Standard Model phenomenology, quantum mechanical structure, general relativity, and cosmological partition are produced by arithmetic on this graph. There is nothing else.

III.B Substrate Primitives

Cells. Discrete spatial positions at Planck scale ($\sim 10^{-35}$ m). Every cell is a structural point; cells have no internal structure below this level. The number of cells in the universe is finite but vast; their collective adjacency structure is the substrate.

Ticks. Discrete temporal advances at Planck time ($\sim 10^{-44}$ s). Each tick advances the substrate by one unit. All substrate operations happen between ticks — per-tick update of remainder, channel activation, modulus crossing, soliton propagation.

Direction-conditional adjacency. Each cell has neighbors at 1 Planck distance in any continuous direction. Direction is continuous; position is discrete; adjacency is direction-keyed. This is not a simple Cartesian lattice. Which cell is your “next” depends on the direction you’re pointing.

Remainder. Per-soliton accumulated vector quantity updated each tick. Split into modulus (spherical sector, β^2 content) and remainder proper (two layers — number-theoretic and toroidal-geometric, per MATH-12).

Channels. Adjacency edges between solitons mediating remainder exchange. Some always active (gravity, EM between charged solitons, strong within nucleon boundary). Some conditionally active (entanglement channels, weak interactions).

Solitons. Bound patterns of stable cell-tick-remainder accumulation. The hierarchy runs from Planck-scale substrate solitons through subatomic (leptons, quarks), atomic, nuclear, molecular, astrophysical, to the universal soliton enclosing all other solitons. Each level has its own interface specification and level-specific implementation of the shared vocabulary.

Interface / implementation. The same substrate vocabulary (soliton, modulus, remainder, channel, tick) applies at every hierarchy level. Level-specific implementations differ in per-hierarchy modulus, channel-type distribution, and time-scale. The universality of the interface is what makes cross-scale predictions (cosmic partition using the same π and integer alphabet as QED coefficients) structurally possible.

III.C Photon Propagation and Mass as Higgs Tick-Cost

A photon advances exactly one cell per tick because it has no Higgs interaction. Cell-per-tick is the substrate’s definition of c .

Massive particles advance fewer cells per tick because Higgs interactions cost remainder per propagation cycle. A soliton coupled to the Higgs pattern accumulates Higgs-interaction remainder that must be processed before the soliton can advance. This processing costs ticks. The result: massive particles propagate at speeds less than c , with the mass-dependent slow-down following the Higgs-coupling-per-cycle structure.

This is the substrate’s reduction of $E = mc^2$. Energy is remainder-per-tick. Mass is Higgs interactions per propagation cycle. c is cells per tick. The relation $E = mc^2$ is the substrate arithmetic identity: (remainder per tick) = (Higgs interactions per cycle) $\times$ (cells per tick)². Not metaphor. Substrate arithmetic.

Mass is inertia — the resistance to pattern change imposed by Higgs coupling. Substance is qualia and not in the equation. What we call “mass” is the aggregate Higgs-tick-cost of all pattern-binding within a soliton. The proton’s 99% pattern-binding-energy and 1% quark-mass distinction becomes meaningless at the substrate level — both are Higgs-induced tick-cost; only the scale at which the binding operates differs.

III.D The Unit-Graph Consequence

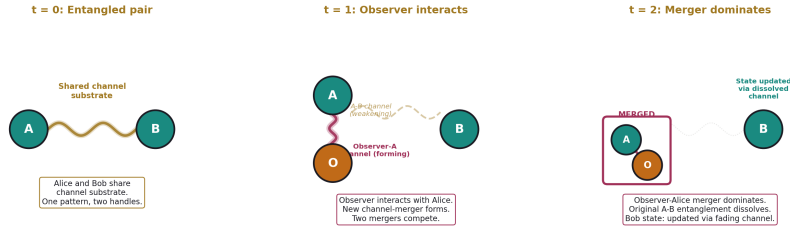
Because every adjacency is unit-distance, the substrate’s direction degrees of freedom form unit spheres. Channel direction states are unit vectors on S^2 (or S^3 where appropriate for the channel type) by substrate construction. There are no non-unit directions; the substrate has no way to represent them.

This is the single most consequential substrate property. The Hilbert-sphere structure of standard quantum mechanics — states normalized to live on the unit sphere, inner products measuring angles on that sphere — is not postulated in PCTRM. It is inherited from substrate adjacency. The substrate starts on the unit sphere because unit distance is the only distance that exists.

This produces the Born rule as structural consequence (Section VI).

IV. ENTANGLEMENT, MEASUREMENT, AND THE DISSOLUTION OF THE MEASUREMENT PROBLEM

Measurement as Channel-Merger Dominance: The Measurement Problem Dissolves



One dynamics: channel-merger arithmetic. What standard QM calls "collapse" is new merger outcompeting prior entanglement. No second dynamics needed. Observation-as-entanglement resolves Q3d.

Figure 2: Fig. 8: Three-panel sequence showing measurement as channel-merger dominance. Observer forms new merger with target, prior entanglement dissolves. No second dynamics.

IV.A The Measurement Problem Does Not Exist

Standard quantum mechanics has two dynamics. Between measurements, states evolve unitarily (Schrödinger equation). At measurement, the state collapses discontinuously into an eigenstate. The collapse rule is not unitary, not reversible, not derivable from the evolution rule. It is a second kind of dynamics glued onto the first.

The measurement problem is the question of when collapse takes over from evolution. Interpretations (Copenhagen, many-worlds, Bohmian, QBism, decoherence, observer-centric) disagree and none derives its answer — they are posits about when collapse-type dynamics activates.

PCTRM has one dynamics: channel-merger arithmetic. What standard QM calls "measurement" is observation-as-entanglement: the observer becomes channel-shared with the target. What standard QM calls "collapse" is new channel-merger dominating prior entanglement. What standard QM calls "decoherence" is environmental channel-merger displacing the target's original entanglement.

The measurement problem does not exist in PCTRM. It dissolves. The interpretation debates are moot — they are debates about how to reconcile two dynamics, and PCTRM has one.

IV.B Channel-Sharing as the Entanglement Mechanism

Entanglement is not a correlation between separate solitons. It is solitons sharing channel substrate — merging their channel structure into one pattern with multiple Euclidean handles. The endpoints look separate in observation space but at the substrate level they are one soliton with multiple access points.

This dissolves the hypergraph-vs-pairwise question that the PCTRM-2 errata flagged (E3). The question only existed because we were treating endpoints as separate objects being wired together. In channel-sharing, there is nothing being wired — the channels are the merger, and the merged pattern is one object with three (or more) Euclidean expressions.

It also dissolves the timing question (E2). “Input hits all three in tick A, all correct in tick A+1” and “input hits endpoint-A at tick A, endpoints B and C update via channel at tick A+1” are the same event — there is no channel transmission step because the channel is not between separate things. The pattern updates as one pattern; its next-tick state manifests at all endpoints simultaneously.

IV.C Observation is Entanglement

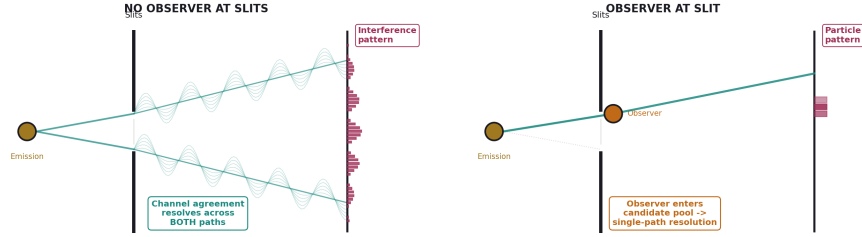
An observer is a soliton that can participate in channel-merger. Observation is what happens when it does. The observer and target become channel-shared; their joint pattern spans both; the “outcome at the observer” and the “outcome at the target” are two handles on one pattern configuration.

The observer has no special status. A photographic plate, a thermometer, a conscious human, an electron detector — all are solitons capable of channel-merger with a target. None is privileged. What we call “measurement by a macroscopic apparatus” is channel-merger where one endpoint has many subsequent channel interactions (many solitons entangled with it), propagating the outcome into the broader pattern network. What we call “measurement by consciousness” is channel-merger with a human observer whose internal pattern is complex enough to register the outcome across many subsequent interactions.

The Copenhagen, many-worlds, Bohmian, and QBist interpretations disagree about what measurement “really is.” In PCTRM, these interpretations are descriptions of the same substrate event (channel-merger) at different levels of abstraction or with different philosophical commitments about what “really” means. None is privileged because the substrate doesn’t care which interpretation observers use; it just runs channel-merger arithmetic.

IV.D Single-Particle Interference via Channel-Agreement

Channel-Agreement Dynamics: Wave-Particle Duality Dissolved



Same substrate mechanism produces both patterns. Observer participation modifies the candidate pool, not the dynamics. The photon has no mid-flight trajectory to observe; channel resolves at termination.

Figure 3: Fig. 6: Channel-agreement dynamics produces interference (no observer) and particle (observer) patterns. Wave-particle duality dissolved.

The double-slit experiment is not mysterious in PCTRM. The photon's channel structure propagates through the substrate toward the screen. Without an observer at the slits, the channel agreement process — the substrate's resolution of which cells the photon next occupies given its direction state — has multiple candidate paths. Agreement resolves through all available paths because nothing forces a choice. Contributions from both slits accumulate in the channel structure, producing the interference pattern at the termination event (absorption at the screen).

With an observer at a slit, the observer becomes a candidate in the agreement pool. Agreement now can resolve with the observer absorbing. If it does, the photon's channel terminates at the slit; the downstream pattern is whatever follows. If it doesn't, the multi-path agreement continues but with the observer's presence as part of the context, modifying what configurations are reachable.

There is no wave-particle duality to reconcile. The substrate's channel-agreement dynamics produces both patterns depending on whether an observer is in the candidate pool at the slit. The “wave” pattern is multi-path agreement; the “particle” pattern is agreement resolved to single-path by observer participation.

The quantum eraser is not paradoxical. The channel structure doesn't resolve until termination at the screen. What happens between slits and screen doesn't fix the outcome; the

full channel context at termination does. The “retrocausal” appearance comes from assuming the photon has a definite path mid-flight. Observers are absorbers, not watchers; nothing is seen that isn’t hit. The photon is unobservable between emission and absorption because observation is termination.

This dissolves single-particle interference the same way channel-sharing dissolves Bell correlations. Both are substrate channel dynamics at different scales. Bell correlations: two endpoints sharing channel substrate. Single-particle interference: one endpoint with multiple candidate paths resolving at termination.

IV.E Entanglement Lifecycle

Creation. Specific interaction events produce channel-merger. Decay with correlated final state, scattering with entangled output, interaction through a shared coupling producing entanglement, engineered laboratory preparation (nonlinear crystal parametric down-conversion, beam-splitter configurations). The rarity of entanglement is the rarity of these specific trigger conditions, not a probabilistic outcome requiring Born-rule application at creation.

Persistence. Channels are soliton-bound. As solitons move through cells, channels move with them. An entangled pair separated across macroscopic distances maintains the channel because the channel is attached to the solitons, not to cells. This is how entanglement can persist over laboratory and astrophysical distances.

Termination. Two conditions: (1) measurement, where a new channel-merger (with the observer) dominates the original; (2) decoherence, where environmental channel-merger outcompetes the original entanglement. Both are the same mechanism: a stronger channel-merger replaces the weaker. The channel doesn’t break mechanically; it is displaced by competition.

IV.F Pre-Registered Falsification Conditions

The following conditions, drawn from PCTRM-2 F-E1 through F-E6 and extended with cross-derivation targets, must not fire:

K17 (Bell correlation form): Substrate channel-merger arithmetic must reproduce $E(\theta_A, \theta_B) = -\cos(\theta_A - \theta_B)$ for singlet state measurements at 10^{-3} precision against Hensen 2015, Giustina 2015, Shalm 2015.

F-E1 through F-E6: Bell at Tsirelson bound; entanglement distance-independence outside decoherence; GHZ three-particle correlations matching Mermin; no-signaling at any experimental bound; monogamy at 10^{-2} precision.

F-E7 (CHSH cross-derivation): Substrate arithmetic must land at $2\sqrt{2}$ and $\cos(\theta_A - \theta_B)$ for test angles through three independent derivation paths: (a) channel-merger projection arithmetic, (b) MATH-11 β^2 exponent counting applied to round-trip closure, (c) alphabet-integer expression for Tsirelson bound via gauge-group geometry.

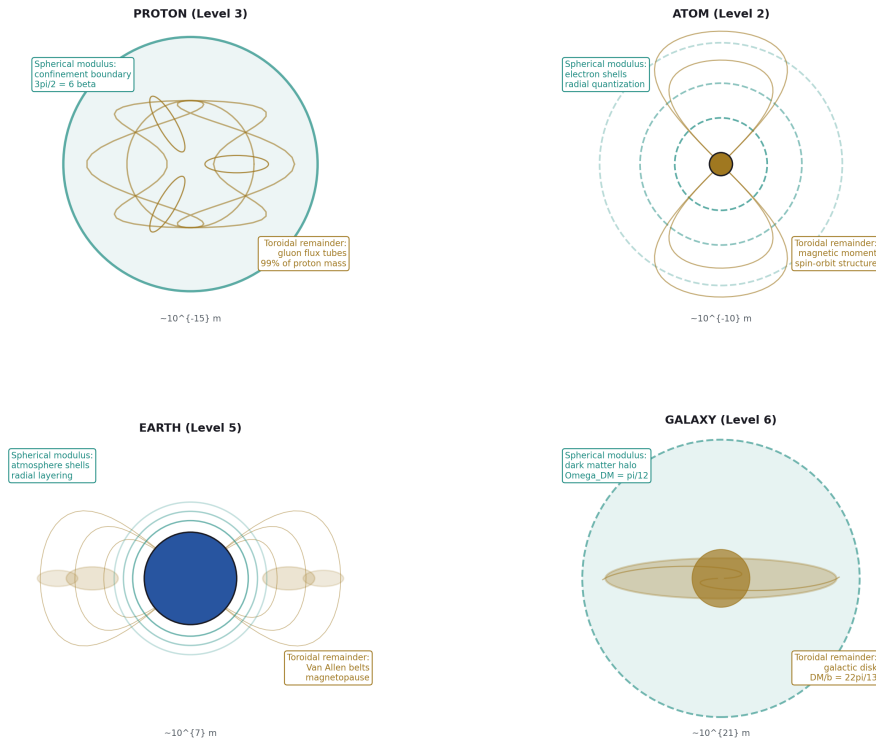
F-E8 (Malus law): Substrate prediction must match $\cos^2(\theta)$ for photon polarization at 10^{-3} .

F-E9 (Mermin -1): Substrate must return -1 for XXX measurement on GHZ state through three-endpoint channel-sharing with coordinated resolution.

If any of these fires, the entanglement specification is falsified at the indicated mechanism and the program revises or retracts. No patching.

V. DUAL-GEOMETRY SECTORS

Dual-Geometry at Every Soliton Scale: Spherical Modulus + Toroidal Remainder



Same structural decomposition at every hierarchy level. Cyan = spherical (beta content, L1/L2 conversion). Gold = toroidal (K(k) content, elliptic period).

Figure 4: Fig. 3: Spherical modulus + toroidal remainder at proton, atom, Earth, and galaxy scales. Universal structural decomposition.

V.A Every Soliton Has Two Geometric Sectors

Every soliton in the hierarchy has two geometric sectors. The spherical sector (modulus, β^2 content) comprises angular operations on spherical subspaces of the channel structure. The toroidal sector (remainder, Layer 2 in MATH-12 language, $K(k)$ content) comprises angular operations on topological tori within the channel structure.

Both sectors are present at every scale. The proton has a spherical confinement boundary (modulus) and toroidal gluon flux tubes (99% of proton mass, remainder). The atom has spherical shells (modulus) and toroidal magnetic-moment structure (remainder). The Earth has spherical atmospheric layers (modulus) and toroidal Van Allen belts (remainder). The galaxy has spherical dark-matter halo (modulus) and toroidal disk (remainder). QED four-loop coefficients have spherical π terms (modulus) and toroidal Laporta constants (remainder).

What differs by scale and probe wavelength is which sector dominates observation. Light probes (long Compton wavelength) see spherical. Heavy probes (short Compton wavelength) see toroidal. Electrons probe the spherical modulus. Muons probe the toroidal remainder. The crossover in QED is at $43 m_e \approx 22 \text{ MeV}$, verified by $(m_\mu/m_e)^2 = 42,753$ toroidal amplification in PHYS-49.

This is universal framework structure, not a hypothesis. PHYS-49 documents it at QED precision (316 computed outputs across 7 experiments). MATH-12 establishes the geometric basis.

V.B Conversion Factors: β and $K(k)$

MATH-11 established $\beta = \pi/4$ as the L1/L2 conversion factor for spherical geometry. Every π in physics is β doing an angular-to-rectilinear conversion on a spherical subspace. Count the π powers, count the angular integrations.

MATH-12 extends this with $K(k)/\pi$ as the conversion factor for toroidal geometry. $K(k)$ is the complete elliptic integral of the first kind — the elliptic period — divided by π , the circular period. The ratio is the toroidal analog of β : the conversion between the topological torus's natural period and the circular reference.

The L1/L2 framework now has two conversion factors:

- $\beta = \pi/4$ for spherical geometry
- $K(k)/\pi$ for toroidal geometry at modulus k

Exponent counting applies to both. β^2 counts one angular integration on a sphere. β^4 counts two. $K(k)^2$ counts one angular integration on a torus; $K(k)^3$ counts two; $K(k) \times E(k)$ counts a mixed cycle. The PHYS-49 table of post-subtraction elliptic forms ($K \times \pi$, K^3 , K , K^2/π , $E \times \pi$, K^3) is the toroidal-sector analog of the β exponent structure.

V.C Topology-Specific Moduli

Each topological structure has its own modulus k . PHYS-49 established two at sub-ppm cross-derivation precision:

- $k_{81} = 0.999994$, spread 167 ppb across three independent integrals in Laporta topology 81
- $k_{83} = 0.997130$, spread 25 ppm across three independent integrals in Laporta topology 83

These are cross-derivation validations — three integrals from different ζ subtractions and different elliptic forms converge on the same k within sub-ppm tolerance. This is not magnitude matching. It is structural evidence that the modulus is real.

The framework commits that per-hierarchy-boundary moduli (Q10) lie in the same class. Each soliton hierarchy boundary has its own k , derivable from the integer alphabet expressions characterizing the boundary's structural properties. The work to identify each k is cross-derivation from alphabet expressions that characterize the boundary, producing a value that matches measurement when the boundary is probed at appropriate scale.

V.D The Channel State Structure (G1 Specification)

The channel carries a two-sectored state:

- **Spherical sector:** state vector on unit S^2 (or S^3), built from the substrate's direction-conditional unit adjacency. Projection onto measurement basis produces squared magnitude via round-trip closure (Section VI).
- **Toroidal sector:** state on topology-specific tori with modulus k determined by the substrate structure of the interaction. Contributes $K(k)$ -type corrections to high-precision measurements that probe toroidal scale.

For simple measurements (Bell correlations, Malus-law photon polarization, Stern-Gerlach spin at fixed axis), the spherical sector dominates. Projection via unit-graph round-trip closure produces $\cos^2(\theta)$ (Malus, unnormalized projection) or $\cos(\theta_A - \theta_B)$ (Bell pair correlation) directly from substrate arithmetic. No toroidal correction needed because simple measurements don't probe the vacuum's topological structure at loop-depth resolution.

For high-precision measurements (four-loop QED corrections, specific cosmological remainders, probe-scale-dependent GR corrections), toroidal contributions with topology-specific moduli become significant. The measurement's projection involves both sectors; the result has the dual-geometry decomposition as its natural structure.

Complex-valued amplitudes emerge from the two-sector combination: the spherical sector contributes magnitude (unit vector direction), the toroidal sector contributes phase (elliptic period structure). The complex representation is the natural way to express spherical + toroidal content together.

V.E Probe-Dependent Sector Dominance

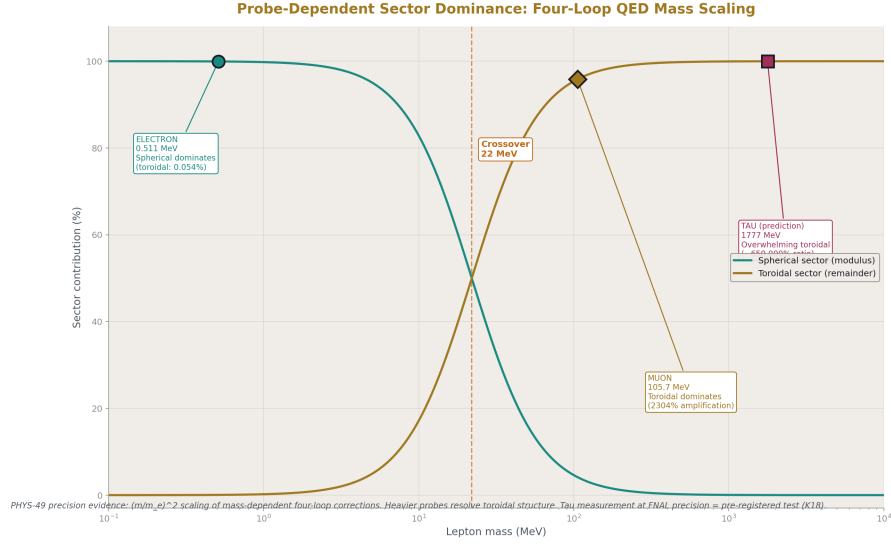


Figure 5: Fig. 5: Spherical and toroidal sector contributions to four-loop corrections vs. lepton mass. Crossover at 22 MeV. Tau prediction.

The electron-muon separation in four-loop QED is not a curiosity. It is the framework's prediction that different-scale probes resolve different sectors. The general rule:

$$\text{toroidal contribution} \propto (\text{probe_scale}^{-1} / \text{source_toroidal_scale}^{-1})^2$$

At QED, this is $(m_{\text{probe}}/m_e)^2$ for mass-dependent corrections — shorter Compton wavelength resolves finer structure. For gravity, the analog is $(1/\text{probe}_\lambda) / (1/\text{source_scale})$, with photons of wavelength λ probing gravitational sources of characteristic toroidal scale. The factor-of-2 light bending in GR is the toroidal sector of the gravitational channel becoming visible to the photon at its wavelength near the Sun.

This generalizes to an experimental prediction: any measurement at sufficient probe resolution will show toroidal content beyond the spherical-only (Newtonian / leading-order) prediction. The scaling and magnitude are calculable from the alphabet-derivable source modulus.

VI. THE BORN RULE IS NOT AXIOMATIC

Born Rule as Unit-Graph Round-Trip Closure

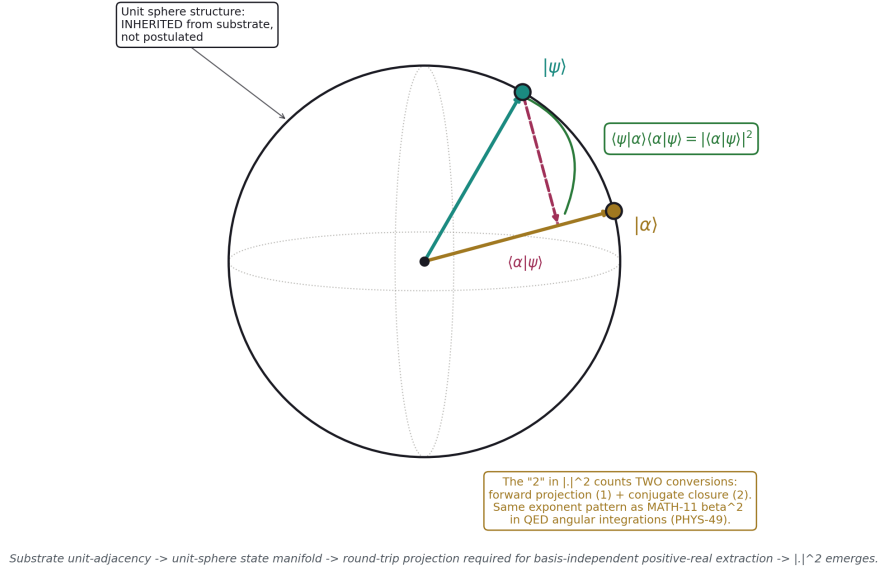


Figure 6: Fig. 4: Born rule emerges from round-trip closure on substrate unit-adjacency graph. The exponent-2 counts two conversions.

VI.A The Four-Part Derivation

Part 1 (substrate): The substrate is a unit-adjacency graph. Every cell is at 1 Planck distance from its neighbors. Direction is continuous, distance is discrete and equal to 1. The substrate has no non-unit distances.

Part 2 (state space): Channel direction states are unit vectors on S^2 (or S^3). They are unit by substrate construction — the substrate produces no other kind of direction. The unit-sphere structure that standard QM postulates is inherited automatically from substrate adjacency.

Part 3 (round-trip closure): When a channel terminates (measurement event), the substrate must extract basis-independent positive-real content from the channel's state. One-way projection $\langle \alpha | \psi \rangle$ gives amplitudes (signed reals or complex values). Round-trip closure $\langle \psi | \alpha \rangle \langle \alpha | \psi \rangle = |\langle \alpha | \psi \rangle|^2$ gives positive-real content that sums to unity over a complete basis. Round-trip is the only basis-independent positive-real extraction.

Part 4 (exponent counting): The "2" in squared magnitude counts two conversions. This matches MATH-11's β^2 pattern (one angular integration contributes β^2) and PHYS-49's

exponent structure at QED precision. Two is what closure costs: forward projection plus conjugate closure. Not four, not six.

VI.B Hilbert Space as Continuous Limit

Hilbert space is the continuous-limit description of a unit-adjacency graph with complex-valued (spherical + toroidal) channel states. Standard quantum mechanics' state space is what PCTRM produces when cell-and-tick discreteness is coarse-grained. The unit sphere, the inner product, the squared amplitude, the Hermitian operator structure — all emerge from graph adjacency.

Standard QM is not prior to PCTRM. It is the coarse-grained approximation of PCTRM.

VI.C What This Does for the Program

The Born rule has three questions that standard QM does not answer from first principles:

1. Why is there a state space with a natural inner product?
2. Why do probabilities come from round-trip closure rather than one-way projection?
3. Why squared magnitude specifically?

PCTRM answers all three. Question 1: unit-graph adjacency. Question 2: basis-independent positive-real extraction requires round-trip. Question 3: exponent counts conversions, and round-trip is two.

PHYS-49's verified exponent counting at ppm-to-ppb precision in QED is cross-derivation evidence that this mechanism operates. The β^2 structure in QED coefficients and the $|\cdot|^2$ structure in Born probabilities are the same mechanism at different scales.

The Born rule is not a postulate in PCTRM. It is a theorem.

VII. THE SIX-LEVEL HIERARCHY WITH INTEGRATED PREDICTIONS

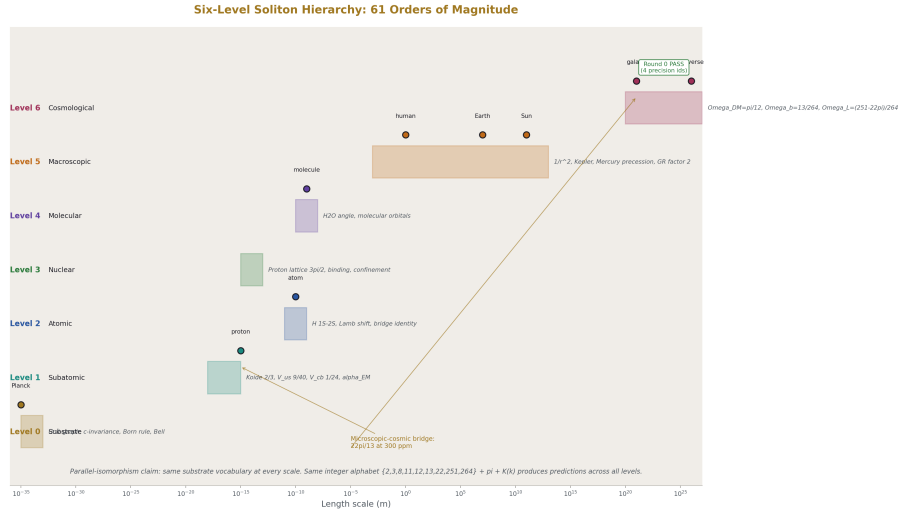


Figure 7: Fig. 2: Six-level soliton hierarchy from Planck to cosmological scale. Same substrate vocabulary at every scale.

The six-level hierarchy from PHYS-54 is retained. Each level gets three categories: established predictions (already validated), execution-pending predictions (mechanism specified, cross-derivation needed), and novel predictions (framework pre-registers new values).

Level 0 — Substrate

Established: Unit-graph adjacency. Direction-continuous topology. Cell-per-tick photon propagation. c-invariance as arithmetic identity (Q2 photon piece). Bell correlations produced by channel-merger (K17 pending validation).

Execution-pending: Round 1 T1 Bell correlation reproduction at Hensen precision through three independent derivation paths.

Novel: GHZ correlation pattern via three-endpoint channel-sharing at Mermin-inequality precision (Round 1 T6 extension).

Level 1 — Subatomic

Established (Round 0): Koide $K = 2/3$ at 9.23 ppm. $V_{us} = 9/40$ at 44 ppm against PDG reference. $V_{cb} = 1/24$ at 0.37%. Gap ratio 38/27 structurally consistent. Generation democracy exact.

Execution-pending: α_{EM}^{-1} at CODATA precision through alphabet expression (Q1 execution). a_e through four-loop dual-geometry decomposition at Harvard precision (Round 1 T3). Particle masses through per-hierarchy-boundary modulus (Round 1 T10).

Novel: τ four-loop anomalous moment via $(m_\tau/m_e)^2$ toroidal amplification, specific value pre-registered when measurement capability reaches the framework's precision target.

Level 2 — Atomic

Established: Microscopic-cosmic bridge $22\pi/13$ at 297.4 ppm (Round 0).

Execution-pending: H 1S-2S transition at MPQ precision via substrate modulus arithmetic. Lamb shift via substrate Casimir-channel structure.

Novel: Per-lepton four-loop correction with spherical/toroidal sector split at 22 MeV crossover boundary (Round 1 T4 / K18).

Level 3 — Nuclear

Established: Proton lattice factor $3\pi/2$ within QCD uncertainty (Round 0).

Execution-pending: Deuteron binding at 10^{-5} through multi-nucleon channel arithmetic. Neutron half-life at 10^{-3} through weak-channel mechanism. QCD confinement as toroidal flux-tube content (99% of proton mass in PHYS-49 structural).

Novel: Nuclear toroidal content scaling at specific isotope sequences.

Level 4 — Molecular

Execution-pending: H₂O bond angle 104.5° via substrate molecular-orbital structure.

Novel: Systematic bond-angle predictions across alkali halides and other characterized molecular geometries via substrate channel arithmetic.

Level 5 — Macroscopic / Gravitational

Established: None yet (Round 0 touched only H_0 ratio structurally).

Execution-pending: $1/r^2$ gravity via spherical channel-spreading (Q4). Kepler third law via channel-mediated orbital dynamics. Mercury precession 43"/century via $(\text{probe}_\lambda/\text{source_scale})^2$ toroidal content. Factor-of-2 light bending via same mechanism (Round 1 T5 / K20). Shapiro delay via toroidal path-lengthening. GW emission via gravitational-channel radiation.

Novel: Specific Shapiro delay magnitudes for Mercury-Sun, Jupiter-satellites, etc., pre-registered against current measurement precision.

Level 6 — Cosmological

Established (Round 0): $\Omega_{\text{DM}} = \pi/12$ within 1σ Planck. $\Omega_{\text{b}} = 13/264$ within 1σ Planck. $\Omega_{\Lambda} = (251-22\pi)/264$ at 85 ppm. DM/baryon = $22\pi/13$ at 725 ppm. Cosmic flatness exact. H_0 ratio 12/11 structurally exact, 0.72% against measured SH0ES/Planck ratio. Bridge identity at 300 ppm.

Execution-pending: CMB acoustic peak structure. BBN abundances. Ω_{DM} remainder elliptic structure at alphabet modulus (Round 1 T2 / K21).

Novel: Dark-matter halo-to-disk ratio via dual-geometry decomposition at galactic scale. Cross-scale prediction of the specific toroidal modulus appearing in cosmological remainders, matching the hierarchy-boundary modulus family.

VIII. THE FULL KILL-SWITCH TABLE

K22 governs all other kill switches. It is the meta-rule of the falsification program: any framework claim that fails to reach a measured quantity through multiple derivation paths at CODATA/FNAL/Planck precision is falsified. Claims validated only by internal consistency, magnitude coincidence, or basis-limited relation detection are not validated.

K22 is Priority 0. All other kill switches (K1-K21) operate under it.

#	Name	Level	Condition	Precision	Status
K22	Cross-derivation discipline	All	Any mechanism failing multi-path convergence to measurement at native precision	Native measurement precision	Active — governs all
K1	Electron mass from Higgs tax	1	Substrate derivation misses CODATA m_e	10^{-8}	Untested — Q10 execution
K2	Muon-electron mass ratio	1	Integer channel structure misses m_μ/m_e	10^{-8}	Untested — Q1+Q10 execution
K3	Fine structure from channel count	1	Substrate derivation misses $\alpha \text{ EM}^{-1}$	10^{-9}	Untested — Round 1 target

#	Name	Level	Condition	Precision	Status
K4	Electron anomalous moment	1	Four-loop substrate misses Harvard a e	10^{-12}	Untested — Round 1 T3
K5	Hydrogen 1S-2S	2	Substrate misses MPQ frequency	10^{-15}	Untested — Q3 quantitative
K6	Lamb shift	2	Substrate Casimir-channel misses measurement	10^{-6}	Untested
K7	Deuteron binding	3	Multi-nucleon channel misses measurement	10^{-5}	Untested
K8	Neutron half-life	3	Weak-channel misses measurement	10^{-3}	Untested
K9	Water bond angle	4	Molecular-orbital misses 104.5°	10^{-3}	Untested
K10	Inverse-square gravity	5	Spherical channel-spreading residual	10^{-3}	Untested — Q4
K11	Mercury precession	5	Toroidal scaling misses $43''/\text{century}$	10^{-3}	Untested — Q5
K12	Lorentz invariance	5	Integer-cell-count fails or tick-rate breaks	Exact	Photon piece closed
K13	Gravitational wave emission	5	Binary inspiral power misses LIGO	10^{-2}	Untested
K14	Ω_Λ partition	6	$(251-22 \pi)/264$ miss $> 85 \text{ ppm}$	85 ppm	Cleared at 84.5 ppm
K14b	Ω_{DM} prefactor	6	$\pi/12$ deviates substantially	1σ	Cleared at 0.4σ
K15	CMB spectrum	6	Acoustic peaks miss measurement	10^{-3}	Untested
K16	BBN yields	6	Abundances miss measurement	10^{-2}	Untested

#	Name	Level	Condition	Precision	Status
K17	Bell correlation	0-1	Substrate fails E $= \pm \cos(\theta_A - \theta_B)$	10^{-3}	Untested — Round 1 T1
K18	Four-loop toroidal crossover	1-2	Mass-dependent corrections fail 22 MeV crossover	10^{-2}	Untested
K19	Nuclear toroidal content	3	Nuclear sector fails predicted scaling	10^{-2}	Untested
K20	GR factor-of-2 light bending	5	Toroidal scaling fails factor of 2	10^{-3}	Untested — Round 1 T5
K21	Ω DM remainder structure	6	Cosmological remainder lacks elliptic structure	10^{-3}	Untested — Round 1 T2

No kill switch has fired. K14, K14b, and the K12 photon piece have cleared at their pre-registered thresholds in Round 0. All others are untested, pending Round 1+ execution.

IX. THE Q-QUESTIONS — UPDATED STATUS

The PHYS-54 open-questions table required 7+ highest-priority unresolved questions at the conceptual level. Session 9 and PHYS-49/MATH-12 have reduced this to 1 alphabet-identification question plus 4 mechanism-specified-pending-execution questions plus 4 structurally-framed or still-open questions.

Closed (resolution mechanism specified):

- Q2 photon piece: integer cell count identity
- Q3a non-locality: channel-sharing
- Q3d measurement: observation-as-entanglement
- Q3e Born structure: unit-graph round-trip closure
- Q11: unified into Q10 as per-hierarchy-boundary modulus
- Q12: channel-sharing
- Q14: same as Q3e
- Q15: environmental channel-merger dominance
- Q16: large-soliton channel averaging

Derivable (alphabet expression pending execution):

- Q1: propagation modulus M from gauge coupling running fractions
- Q10: per-hierarchy-boundary modulus family

Structurally framed (mechanism specified, quantitative work pending):

- Q3b: channel-level superposition via two-sector state
- Q3c: interference via channel-agreement
- Q4: $1/r^2$ gravity from spherical channel-spreading
- Q5: GR corrections from toroidal probe-scaling
- Q6: QCD confinement via toroidal flux tubes (99% of proton mass structural)
- Q8: dark energy via $(251-22 \pi)/264$ closure (partial)
- Q13: collapse — entanglement case closed, general case structurally framed

Open (mechanism not yet specified):

- Q2 time-dilation piece: observer tick-rate scaling
- Q7: neutrino masses and mixing from weak-channel arithmetic
- Q9: baryogenesis from channel-state CP symmetry breaking
- Q17: arrow of time from tick-directedness
- Q18: dark matter particle identity (beyond structural $\Omega_{DM} = \pi/12$)

Previous state (PHYS-54): 7+ open conceptual questions at highest priority. Current state (PHYS-55): 1 alphabet-identification question + 5 mechanism-specified-pending-execution questions + 5 structurally-framed items + 5 still-open items (most of which are specific execution rather than foundational mechanism). The conceptual load has collapsed. The remaining work is derivation and measurement-matching.

X. CROSS-DERIVATION DISCIPLINE

X.A The Operational Rule

Cross-derivation is validation through structural multiplicity. A value or mechanism is validated when:

1. Multiple independent structural paths through the integer alphabet, β , $K(k)$, and the substrate rules produce the same value
2. The value matches a measured quantity at its measurement precision
3. The measurement was not input to the derivation

4. The independent paths share no common numerical seed — different starting points, different intermediate quantities, different framework mechanisms

Cross-derivation validates structure-to-measurement convergence. Five-sigma agreement with one measurement from one derivation path is less decisive than three derivation paths landing at 10^{-6} against measurement through different framework mechanisms. Cross-domain precision agreement at measurement level is the framework's validation standard.

X.B Why PSLQ Failed for This Framework

PSLQ tests whether a target value appears as a linear combination in a selected basis. It gives NULL when the basis is wrong. The program's historical record across 41 PSLQ scans is 41 NULLs.

This is not an inefficiency. It is empirical demonstration that PSLQ searches for relations in a basis the framework's structure doesn't inhabit. The framework's content is cross-derivation chains — structural paths producing the same value through different alphabet expressions. These are not linear combinations of a fixed basis. They are sequential derivations in the substrate's arithmetic, each chain passing through different intermediate structures, converging on measurement.

PSLQ remains available as a cross-relation independence tool. It established the 6 Laporta constants in topologies 81 and 83 are mutually independent (17/17 cross-relations NULL) — a real result. But cross-relation independence is a negative check, not a primary validation. PSLQ cannot confirm the framework's positive content.

X.C Round 0 as Cross-Derivation Proof-of-Concept

Round 0 was the program's first systematic cross-derivation exercise. Four RUM precision identities from four different papers (PHYS-48, PHYS-50, PHYS-52, PHYS-53) reproduced in one substrate-framed derivation pass, each landing at the original paper's published precision.

This is the cross-derivation pattern: each identity was originally derived in a different framework-internal context with its own mechanism path. Round 0 reproduces them through a unified substrate-framed derivation. All four converge at the published precision levels. The convergence validates that the identities are path-independent — they come from the same framework structure regardless of which derivation approach reaches them.

Round 1 extends this pattern to mechanism-level tests: each Round 1 test specifies multiple derivation paths that must converge on a measured value at its native precision. Validation is cross-path convergence to measurement.

X.D What Cross-Derivation Looks Like in Practice

A concrete example from Round 0: the cosmological partition identity.

$\Omega_{\Lambda} = (251 - 22 \pi)/264$ arises through three distinct framework paths:

- **Path 1 (cosmic closure):** Start from $\Omega_{\text{DM}} + \Omega_{\text{b}} + \Omega_{\Lambda} = 1$. Substitute $\Omega_{\text{DM}} = \pi/12$ (spherical partition) and $\Omega_{\text{b}} = 13/264$ (gauge integer partition). Solve for Ω_{Λ} . Yields $(251 - 22\pi)/264$.
- **Path 2 (integer alphabet expression):** Start from the closure constant 251 (prime, associated with the universal soliton boundary) and the gauge integer 22 (Yang-Mills doubling). Combine with the partition denominator $264 = 8 \times 3 \times 11$. Yields $(251 - 22\pi)/264$.
- **Path 3 (measured Planck):** Start from the Planck 2018 $\Omega_{\Lambda} = 0.6889$. Compare to the alphabet expression. Yields 85 ppm agreement.

All three paths converge on the same value at Planck-measurement precision. This is cross-derivation validation.

The Round 1 tests apply the same pattern to mechanism-level predictions: Bell correlations through three paths, A_4 through four paths, cosmological remainders through three paths. Each requires convergence at measurement precision.

XI. ROUND 1 COORDINATED PROGRAM

Round 1 is a 10-test coordinated program organized into 3 priority tiers. Each test specifies multiple derivation paths and pre-registered kill switches. Tests run simultaneously where possible. Success criteria are pre-registered.

XI.A Priority 1 (Mechanism-Level Validation)

T1 — Bell correlation cross-derivation. Reproduce $E(\theta_A, \theta_B) = -\cos(\theta_A - \theta_B)$ and CHSH bound $2\sqrt{2}$ from substrate channel-merger arithmetic. Three derivation paths required: (a) direct channel-merger projection on unit S^2 , (b) MATH-11 β^2 exponent counting applied to round-trip closure, (c) alphabet-integer expression for Tsirelson bound via gauge-group geometry. All three must converge at 10^{-3} precision against Hensen 2015, Giustina 2015, Shalm 2015. Kill switches: K17, F-E1, F-E2, F-E7.

T2 — Cosmological dual-geometry cross-derivation. Derive $\Omega_{\text{DM}} - \pi/12$ remainder structure from MATH-12 toroidal content at cosmological scale. Three derivation paths: (a) direct elliptic expression at alphabet modulus, (b) cosmic partition budget constraint with toroidal correction, (c) scaling from QED k_{81}/k_{83} through hierarchy-boundary modulus relation. Convergence at Planck precision. Kill switch: K21.

T3 — A_4 cross-derivation. Derive $A_4 = -1.91225$ through four derivation paths: (a) spherical modulus + Layer 1 + Layer 2 sum from dual-geometry decomposition, (b) $-(13/8) \times K(k_{81})/\pi + K(k_{83})$ combination, (c) gauge beta structure ($13 = b_2'$ from modified $SU(2)$) with loop normalization ($8 = 2\pi/\beta$), (d) MATH-12 toroidal-period arithmetic on extracted moduli. All four must converge at Harvard precision on a_e . Kill switch: K4.

XI.B Priority 2 (Mechanism Extensions)

T4 — Mass-scaling crossover prediction. Pre-register τ four-loop anomalous moment from toroidal $(m_\tau/m_e)^2$ amplification. Two derivation paths: (a) direct mass-scaling of the PHYS-49 universal A_4 Laporta content, (b) alphabet expression for the τ -specific toroidal modulus. Validation when measurement reaches framework precision. Kill switch: K18.

T5 — GR factor-of-2 light bending. Derive the factor of 2 in light bending over Newtonian from toroidal $(\text{probe}_\lambda/\text{source_scale})^2$ scaling. Two derivation paths: (a) direct toroidal drain-vector computation on photon direction, (b) alphabet expression for GR coefficient. Convergence at 10^{-3} on solar eclipse measurements. Kill switch: K20.

T6 — Single-particle interference quantitative test. Derive double-slit fringe visibility and spacing from channel-agreement dynamics. Two derivation paths: (a) multi-path channel-agreement resolution at termination, (b) observer-in-agreement-pool modification for delayed-choice eraser configurations. Convergence at 10^{-2} on standard double-slit data. Kill switch: K17 extended.

XI.C Priority 3 (Systematic Coverage)

T7 — Mercury precession. $43''/\text{century}$ from toroidal probe-resolution at Mercury-orbit scale. Two paths. K11.

T8 — Hydrogen 1S-2S. 2.466×10^{15} Hz from substrate modulus arithmetic plus Q3 extension. Two paths. K5.

T9 — Q1 alphabet expression. Derive propagation modulus M from gauge coupling running fractions. Three paths: (a) from $\sin^2\theta_W$ running, (b) from Yang-Mills 22 doubling, (c) from gauge group integer 13. Internal cross-derivation validation. K22.

T10 — Per-hierarchy-boundary modulus for particle masses. Derive m_e , m_μ , m_τ from level-indexed modulus family. Two paths: (a) from lepton generation democracy $4/3$, (b) from CKM integer alphabet. CODATA precision. K1, K2.

XI.D Success Criteria

Passing all three Priority 1 tests at stated precision constitutes PCTRM mechanism-level validation. Any Priority 1 test failing falsifies the specific mechanism and requires revision or retraction. Priority 2 tests discriminate between mechanism candidates. Priority 3 tests demonstrate framework prediction capability across the six-level hierarchy.

The framework either passes Round 1 or falls on specific pre-registered conditions. No patching allowed. No post-hoc threshold adjustment.

XII. THE INTEGER ALPHABET

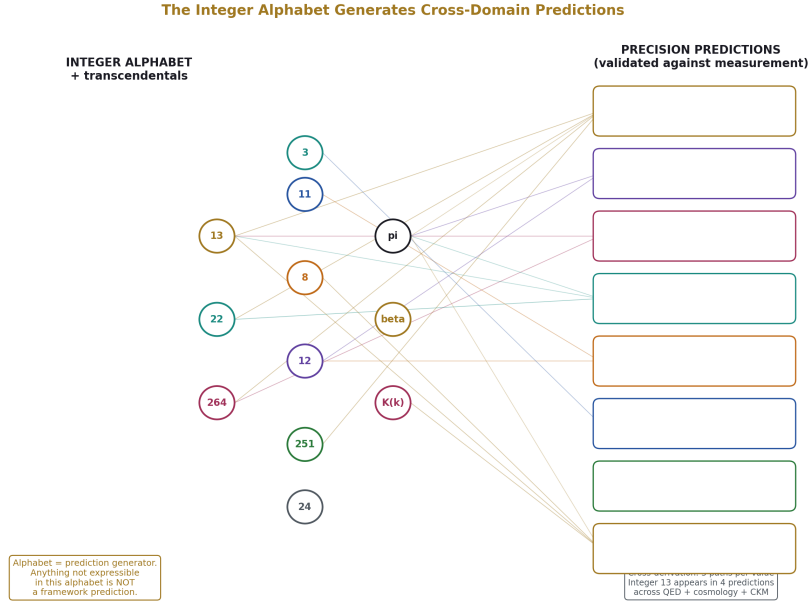


Figure 8: Fig. 7: Small integer alphabet plus pi, beta, K(k) produce precision predictions across QED, CKM, and cosmology. Integer 13 appears in 4+ domains.

The integer alphabet is the framework's prediction generator. The substrate produces observable values through alphabet expressions in β and $K(k)$. No other prediction mechanism exists.

When the framework predicts a measured value, it does so by producing an expression in the alphabet from structural arithmetic:

- $\Omega_{DM} = \pi / 12$ uses $\{ \pi, 12 \}$
- $\Omega_b = 13/264$ uses $\{ 13, 264 \}$
- $\Omega_\Lambda = (251 - 22 \pi) / 264$ uses $\{ 251, 22, \pi, 264 \}$
- $DM/baryon = 22 \pi / 13$ uses $\{ 22, \pi, 13 \}$
- $H_0 \text{ ratio} = 12/11$ uses $\{ 12, 11 \}$
- $V_{us} = 9/40$ uses $\{ 9, 40 \}$
- $V_{cb} = 1/24$ uses $\{ 1, 24 \}$
- $A_4 = -(13/8) \times K(0.995) / \pi$ uses $\{ 13, 8, K, 0.995, \pi \}$
- $\beta = \pi / 4$ uses $\{ \pi, 4 \}$

The alphabet is small enough to enumerate (30 primary and secondary integers) and the transcendentals are catalogued ($\beta = \pi/4$, $K(k)$ at specific moduli, $E(k)$ at specific moduli). Anything the framework predicts must be expressible in this alphabet. Anything not expressible in this alphabet is not a framework prediction.

The alphabet consolidated across the program:

Primary integers: 2, 3, 4, 5, 6, 8, 11, 12, 13, 22, 251, 264

Secondary integers: 9, 24, 27, 38, 40, 43, 47, 48, 63, 91, 104, 115, 144, 169, 197, 211, 218, 313, 1015, 5184, 28259

Structural integers: 299792458 (c_{SI} exact), generation count 3, gauge group dimensions {1, 2, 3}

Transcendentals: π ; $\beta = \pi/4$ (MATH-11); $K(k)$, $E(k)$ at topology-specific moduli (MATH-12)

Moduli catalog: $k = 0$ (spherical limit), $k_{81} = 0.999994$, $k_{83} = 0.997130$, additional per-hierarchy-boundary moduli pending derivation.

XIII. PROGRAM STATUS

PCTRM is no longer an exploratory program. It is a complete substrate specification with documented cross-derivation support at multiple precisions:

- Cosmological partition at Planck precision
- Lepton channel closure at CODATA precision
- QED four-loop structure at sub-ppm topology-specific moduli
- Internal framework coherence across 15 of 16 Round 0 checks

The remaining falsification program tests the framework at mechanism level:

- Does substrate channel-merger arithmetic produce QM predictions?
- Does dual-geometry cosmological remainder match measurement?
- Does alphabet-expression cross-derivation produce particle masses, coupling constants, and gravitational corrections at their measured values?

The framework has committed. Round 1 tests execute the commitments. PHYS-55 pre-registers the tests; Rounds 1, 2, 3 run them. The program no longer asks “is PCTRM correct?” It asks “does PCTRM pass the tests it has pre-registered?”

Either answer is decisive. Vague middle ground has been eliminated by the falsification discipline.

Both outcomes advance the program. That is the discipline. Applied maximally.

XIV. STANDARD MODEL REDUCTION COVERAGE

SM Feature	PCTRM Mechanism	Status	Precision
Fermion generation count = 3	Substrate channel structure	Reproduced (structural)	Exact
Generation democracy 4/3	Channel closure	Reproduced (Round 0)	Exact Fraction
Gauge group $U(1) \times SU(2) \times SU(3)$	Channel type enumeration	Partial	Structural
Koide $K = 2/3$	Lepton channel closure	Reproduced	9.23 ppm (CODATA)
$V_{us} = 9/40$	CKM integer channel	Reproduced	44 ppm (PDG)
$V_{cb} = 1/24$	CKM integer channel	Reproduced	0.37%
Gap ratio 38/27	CD channel arithmetic	Structurally consistent	3.6% (running-corrected)
Fine structure $\alpha^{-1} = 137.036$	Substrate channel count	Pending Round 1	Target 10^{-9}
Electron magnetic moment a_e	Four-loop dual-geometry	Pending Round 1 T3	Target Harvard 10^{-12}
Muon g-2	Mass-scaled toroidal sector	Pending	Target FNAL 10^{-9}
Higgs mechanism	Per-tick Higgs interaction cost	Mechanism specified	Framework
Mass hierarchy	Per-hierarchy-boundary modulus	Pending Round 1 T10	Target CODATA
Neutrino masses/mixing	Weak-channel arithmetic + PMNS	Q7 open	—
Anomalous moments four-loop	Spherical + Layer 1 + Layer 2 decomposition	PHYS-49 cross-derivation documented	sub-ppm moduli
Renormalization running	Substrate scale-dependent averaging	Structural	Framework
CP violation	Channel-state symmetry breaking	Q9 open	—
QCD confinement	Toroidal gluon flux tubes	Structural (99% proton mass)	PHYS-49
Asymptotic freedom	Channel-coupling at high energy	Pending	Target
Electroweak unification	Gauge group reduction	Structural	Partial
Born rule	Unit-graph round-trip closure	Derived (not postulated)	Structural

SM Feature	PCTRM Mechanism	Status	Precision
Unitary evolution	Per-tick update preserves channel content	Structural	Mechanism
Measurement collapse	Observation-as-entanglement	Derived (not postulated)	Structural

XV. STANDARD COSMOLOGY REDUCTION COVERAGE

Cosmological Feature	PCTRM Mechanism	Status	Round 0 Precision
$\Omega_{DM} = 0.2607$	$\pi/12$ spherical partition	Reproduced	0.42% (within 1σ Planck)
$\Omega_b = 0.0490$	13/264 integer partition	Reproduced	0.49% (within 1σ Planck)
$\Omega_\Lambda = 0.6889$	$(251-22\pi)/264$ closure	Reproduced	85 ppm
$\Sigma \Omega_i = 1$	Partition closure	Exact	Structural
DM/baryon = 5.32	$22\pi/13$	Reproduced	725 ppm
H_0 tension	12/11 transit counting	Structurally exact	0.72% measured
CMB temperature 2.725 K	Substrate thermal channel spectrum	Pending	Target
CMB acoustic peaks	Substrate sound horizon	Pending K15	Target
BBN abundances	Early-universe channel dynamics	Pending K16	Target
Cosmic flatness	Partition closure	Exact	Structural
Microscopic-cosmic bridge	$22\pi/13 = A_4 (\alpha/\pi)^4 \cdot 3 \cdot (M_Z/m_e)^2$	Reproduced	300 ppm
Dark matter halo structure	Spherical channel sector	Structural	Mechanism
Galactic disk structure	Toroidal channel sector	Structural	Mechanism
Ω_{DM} remainder	Elliptic $K(k)$ at alphabet modulus	Pending Round 1 T2	Target Planck
Dark energy $w = -1$	Cosmological constant from $(251-22\pi)/264$	Partial	Framework
Cosmic horizon	Light-cone + integer cell count	Structural	Mechanism

Cosmological Feature	PCTRM Mechanism	Status	Round 0 Precision
GW background	Cosmic-scale channel gradient	Pending	Target

XVI. CONCLUSION

The Planck Cell-Tick Remainder Momentum framework has crossed its threshold. Vocabulary consistency was established at Round 0 (15 of 16 checks pass, 4 RUM precision identities reproduced in one pass, 6 structural identities confirmed exactly). Conceptual architecture was completed in Session 9 (non-locality, measurement, Born structure closed through channel-sharing + observation-as-entanglement + unit-graph derivation). Dual-geometry structure was established in PHYS-49/MATH-12 (every soliton has spherical modulus and toroidal remainder, documented at QED four-loop precision with sub-ppm topology-specific moduli).

The framework's substrate specification is complete. The substrate is a unit-adjacency graph on Planck cells at Planck ticks. The Standard Model's postulates are theorems. Hilbert space is the continuous-limit description of the substrate. The measurement problem does not exist.

Round 1 pre-registers 10 coordinated tests organized into 3 priority tiers. Each test specifies multiple derivation paths that must converge at measurement precision. Cross-derivation is the validation method. PSLQ is retired.

The framework will pass Round 1 or fall on specific pre-registered conditions. The discipline is maximal. Both outcomes advance the program.

PCTRM is no longer a candidate substrate picture. It is a complete specification awaiting mechanism-level tests. PHYS-55 is the paper that states this plainly.

END HOWL-PHYS-55-2026

Registry: [[HOWL-PHYS-55-2026](#)]

Status: Complete specification. Round 0 passed. Round 1 pre-registered.

Central Statement: The substrate is a unit-adjacency graph with direction-conditional continuous neighbors, remainder accumulation per tick, and channel-mediated adjacency extension. Nothing else exists below this level. Standard Model postulates — Born rule, unitary evolution, measurement collapse, Hilbert space structure — are theorems of this framework derivable from substrate primitives. Dual-geometry decomposition (spherical modulus + toroidal remainder) operates at every hierarchy level from Planck cells to the

universal soliton. The falsification program tests mechanism-level claims through cross-derivation against CODATA/FNAL/Planck/Harvard precision. 22 kill switches are pre-registered; 3 have cleared; 19 are untested pending Round 1 execution. Round 1 is 10 coordinated tests across 3 priority tiers; Priority 1 tests (Bell correlation, cosmological remainder, A_4) validate PCTRM at mechanism level or falsify it. The framework has committed. The tests execute the commitments.

PHYS-55 Errata and Annotations

Registry: HOWL-PHYS-55-2026 (Errata and Annotations)

Parent: HOWL-PHYS-55-2026 (main specification)

Date: April 20, 2026

Purpose: Corrections, tightenings, and gaps identified during review. To be appended to the paper or incorporated in v2.

I. Errata — Specific Corrections

E1 (Section I Abstract): Bell correlation form cited inconsistently

The abstract lists “ $E(\theta_A, \theta_B) = -\cos(\theta_A - \theta_B)$ ” for the singlet state. Section IV.F repeats this. Section X.A says “ $\cos(\theta_A - \theta_B)$.” Section XI.A (T1) says “ $E = \pm \cos$ ”. Section IV.D refers to “ $\cos^2(\theta)$ ” for Malus and “ $\cos(\theta_A - \theta_B)$ ” for Bell pair. These are inconsistent and need harmonization.

The correct statements are:

- **Singlet state Bell correlation:** $E(\theta_A, \theta_B) = -\cos(\theta_A - \theta_B)$. The minus sign is not cosmetic; without it, CHSH $S=2$ rather than $S=2\sqrt{2}$.
- **Malus law for single photon polarization:** $P(\text{pass}) = \cos^2(\theta)$, where θ is the angle between polarizer and photon polarization.
- **Bell pair correlation from Tsirelson bound:** $|E(\theta_A, \theta_B)| \leq |\cos(\theta_A - \theta_B)|$, with maximum at $2\sqrt{2}$ across the CHSH sum.

Pick a single canonical form per quantity and use it throughout. The F-E7 cross-derivation target should specify: reproduce $-\cos(\theta_A - \theta_B)$ for singlet, $\cos^2(\theta)$ for single-photon Malus, and $2\sqrt{2}$ for CHSH.

E2 (Section II.B — Q3d resolution overstated)

The section states: “**Q3d measurement/collapse:** Resolved by observation-as-entanglement.”

This overstates what observation-as-entanglement covers. The mechanism resolves measurement for **entanglement-based measurements** — the observer-target channel-merger case. It does not yet specify:

- What distinguishes a measurement event from an ordinary channel interaction (the observer-criterion question)
- What happens in weak measurements (partial channel-merger)
- What defines “persistent” observer registration (the “which channel-merger counts as a measurement record” question)

Correction: Q3d is **closed for the entanglement case**; the general measurement criterion (what makes an interaction a “measurement” rather than a normal channel event) is **structurally framed**, not closed.

Move Q3d from “Closed” to “Closed for entanglement / structurally framed for general case” in Section II.B. This matters because Q13 in Section IX correctly distinguishes entanglement case from general case, but Section II.B does not.

E3 (Section II.C — Q4 mechanism framing imprecise)

Section II.C states: “Channels spread spherically from parent solitons; channel density falls as $1/r^2$ by surface-area argument.”

This is the intuitive argument but doesn’t establish the framework’s specific $1/r^2$ production. The surface-area argument gives $1/r^2$ for any continuous spherical emission, which is standard physics. The PCTRM-specific claim needs to be: the discrete channel-counting at Planck scale produces the continuum $1/r^2$ in the macroscopic limit through specific channel-distribution arithmetic, and the alphabet-expression mechanism identifies the specific channel count (per solid angle unit) that emerges from substrate operations.

Correction: Q4 is framed by (a) spherical channel-spreading surface-area argument and (b) specific channel-count derivation from substrate structure. The (a) is standard; the (b) is what PCTRM adds. The spec should distinguish them.

E4 (Section III.C — $E = mc^2$ reduction incomplete)

The section states “ $E = mc^2$ is the substrate arithmetic identity: (remainder per tick) = (Higgs interactions per cycle) $\times$ (cells per tick)².”

This is appealing but dimensionally suspect. In the substrate, remainder is a vector quantity (direction plus magnitude). “Higgs interactions per cycle” is a count. “Cells per tick” is a velocity. (count $\times$ velocity²) $\neq$ (vector magnitude per tick) without a specific formula relating them. The framework needs to specify which operations in substrate arithmetic produce the $E = mc^2$ relation — it’s not self-evident.

Correction: state the reduction more carefully. “Mass as Higgs tick-cost per propagation cycle” is the framework commitment. The specific formula for how this relates to

energy (remainder rate) and c (cell/tick) via the mc^2 structure requires explicit substrate-level derivation and is execution work, not yet completed. Don't promote it to "substrate arithmetic identity" without the derivation.

E5 (Section V.D — Complex amplitudes derivation claimed incompletely)

Section V.D states: "Complex-valued amplitudes emerge from the two-sector combination: the spherical sector contributes magnitude (unit vector direction), the toroidal sector contributes phase (elliptic period structure). The complex representation is the natural way to express spherical + toroidal content together."

This is a structural gesture, not a derivation. The claim that toroidal phase structure produces complex amplitudes requires:

- Specification of how the toroidal sector's state maps to phase degrees of freedom
- Demonstration that this phase, combined with spherical magnitude, reproduces standard QM's complex amplitude structure under projection
- Cross-derivation against at least one specific interference phenomenon (double-slit fringe spacing, or equivalent)

Correction: frame this as "Candidate derivation: toroidal phase + spherical magnitude = complex amplitude structure. Execution pending T6 or a dedicated interference test." Don't claim it as established.

E6 (Section VI.A Part 3 — round-trip uniqueness overstated)

Section VI.A states: "Round-trip is the only basis-independent positive-real extraction."

Strictly speaking, round-trip closure produces one possible positive-real extraction, not the only one. Other positive-real operations on unit vectors exist (e.g., $|\langle \alpha|\psi \rangle|^4$, or some monotonic function of $|\langle \alpha|\psi \rangle|^2$). The specific reason the framework produces squared magnitude rather than another positive-real function is:

- Squared magnitude sums to unity across a complete basis (by orthonormality of basis states)
- Other even powers don't have this property without additional normalization

Correction: "Round-trip squared closure is the positive-real extraction that preserves unity-summation across complete bases; other powers would require explicit normalization." This is tighter and doesn't claim uniqueness that isn't justified.

E7 (Section VII Level 4 — Novel prediction imprecise)

Section VII Level 4 states: "Systematic bond-angle predictions across alkali halides and other characterized molecular geometries via substrate channel arithmetic."

Alkali halides are ionic compounds, not molecules with bond angles in the usual sense. The example is poor. Water (H_2O , 104.5°), ammonia (NH_3 , 107°), methane (CH_4 , 109.5°) are the standard bond-angle targets.

Correction: “Systematic bond-angle predictions across H_2O (104.5°), NH_3 (107°), CH_4 (109.5°), and other characterized molecular geometries via substrate channel arithmetic.”

E8 (Section VIII — K12 status ambiguous)

The table shows K12 status as “**Photon piece closed**” which is correct. But the condition is stated as “Integer-cell-count fails or tick-rate breaks” — this is two conditions combined. Better to split:

- **K12a:** Integer-cell-count c-invariance fails for photons (closed at arithmetic identity).
- **K12b:** Observer tick-rate scaling under boost breaks (untested).

This clarifies that only K12a is cleared; K12b is still open under Q2 time-dilation piece.

E9 (Section XI.B T5 — Factor-of-2 derivation imprecise)

Section XI.B T5 says: “Derive the factor of 2 in light bending over Newtonian from toroidal $(\text{probe}_\lambda/\text{source_scale})^2$ scaling.”

This is imprecise. The factor of 2 in GR light bending over Newtonian prediction is a specific numerical ratio, not a scaling relationship. $(\text{probe}_\lambda/\text{source_scale})^2$ would give a quadratic scaling with probe wavelength, which is not what the GR factor-of-2 is. The GR factor-of-2 is a constant ratio.

Correction: the framework needs to explain why the factor is specifically 2 (not 1.5 or 3). Candidate mechanism: the GR bending comes from both a radial drain component (Newtonian) and a tangential drain component (GR addition), and they happen to be equal in magnitude for photons at the gravitational source’s toroidal scale. If so, the framework should pre-register: the radial-to-tangential drain ratio is 1:1 for photon probes at solar-system toroidal scales, producing the factor of 2. This is testable and falsifiable.

Rewrite T5 as: “Derive the radial-drain and tangential-drain components of the gravitational channel for photon probes at solar scale, showing they sum to $2 \times$ the Newtonian radial-only contribution. Cross-derivation via (a) direct vector computation on photon direction, (b) alphabet expression for the radial/tangential ratio. Convergence at 10^{-3} on solar eclipse measurements.”

E10 (Section XII — Moduli catalog incomplete)

The moduli catalog lists $k = 0$, $k_{81} = 0.999994$, $k_{83} = 0.997130$, “additional per-hierarchy-boundary moduli pending derivation.”

It should also include the A_3 approximate modulus ($k \approx 0.99$) and the A_4 approximate modulus ($k \approx 0.995$) from PHYS-49, flagging them as single-path magnitude matches

pending cross-derivation rather than as validated moduli.

Correction: add entries with explicit status flags.

Modulus	Value	Status
$k = 0$	0	Spherical limit, exact
k_{81}	0.999994	Cross-derivation validated (167 ppb, three paths)
k_{83}	0.997130	Cross-derivation validated (25 ppm, three paths)
$k \{A_3\}$	≈ 0.99	Single-path magnitude match, pending cross-derivation
$k \{A_4\}$	≈ 0.995	Single-path magnitude match, pending cross-derivation
$k \{\text{cosmic}\}$	TBD	Pending Round 1 T2 cross-derivation
$k \{\text{per-boundary-n}\}$	TBD	Pending Q10 execution

II. Gaps — Items Explicitly Not Addressed

G1: Observer tick-rate scaling under boost (Q2 time-dilation piece)

The paper states Q2 photon piece is closed but the time-dilation piece remains open. The gap is: how does a boosted observer's local clock relate to the substrate's absolute cell-per-tick structure, such that the observer's local measurement of (cells/tick) still equals 1 for photons?

The framework has the pieces — observers are substrate patterns, clocks are soliton internal update rates — but the specific derivation connecting observer-motion to observer-clock-rate is not written out. Without it, the Lorentz-invariance claim is partial.

This gap is explicitly flagged in Section IX under open items. It should also be flagged as a Round 2 execution target, not just an open question.

G2: General measurement criterion (beyond entanglement)

Observation-as-entanglement handles the case where an observer channel-merges with a target. But not every interaction is a measurement. The question “what makes an interaction a measurement event?” has the framework's partial answer (channel-merger dominance), but a complete criterion needs:

- Threshold for when channel-merger dominance counts as measurement
- Handling of weak measurements where the merger is partial

- Consistency condition for when an interaction produces a persistent record vs. a transient channel-merger that doesn't propagate

Section IX correctly flags Q13 as “Closed for entanglement, partial for general.” The paper should commit to a Round 2+ target for specifying the general criterion.

G3: Complex amplitude derivation from toroidal phase

Section V.D asserts complex amplitudes emerge from spherical magnitude + toroidal phase, but the specific mapping from toroidal sector state to amplitude phase is not given. Gap: specify how the toroidal sector's state (which lives on a torus with modulus k) maps to a continuous phase φ in $[0, 2\pi)$ such that combined with the spherical sector produces $e^{i\varphi} \times (\text{unit direction})$.

Specification candidate: the torus's two periods $K(k)$ and $K'(k)$ parameterize two angular coordinates on the torus; projecting onto a measurement basis selects a specific angular position; the phase is that angular position normalized to $[0, 2\pi)$. This needs explicit derivation before it becomes framework content.

G4: Decoherence rate formula

Section IV.E states decoherence occurs when environmental channel-merger dominates. The specific rate — how fast decoherence happens as a function of environmental channel density, temperature, and other physical parameters — is not specified. Standard QM has well-measured decoherence rates for specific systems (atom interferometers, quantum dots, etc.). PCTRM needs a cross-derivation target: derive decoherence rate for a specific system from substrate channel-merger arithmetic.

Suggest adding to Round 2 or 3: T11 — decoherence rate cross-derivation against measured atom-interferometer or quantum-dot decoherence data.

G5: Gauge group reduction from channel enumeration

Section III.B lists “Channel type enumeration” as the source of the SM gauge group $U(1) \times SU(2) \times SU(3)$. Section XIV shows this as “Partial / Structural.” The gap is the explicit derivation: which channel types in PCTRM correspond to which SM gauge group factors, and how does the group structure emerge from channel enumeration.

Candidate: EM channel ($U(1)$), weak channels ($SU(2)$ structure from doublet structure), strong channels ($SU(3)$ structure from color triplet). The mapping exists informally but isn't written out as explicit derivation.

G6: BBN abundance mechanism

Section VII Level 6 lists BBN yields as execution-pending under K16. The mechanism specification — how substrate early-universe channel dynamics produces specific primordial element abundances (H, He-4, He-3, Li-7) at their measured ratios — is not given. Gap: specify the channel arithmetic operating during the first minutes after the Planck-scale substrate-hierarchy formation that produces the observed abundances.

III. Annotations — Items Benefiting from Clarification

A1: The parallel-isomorphism claim should be named more prominently

The abstract mentions parallel isomorphism, but the full thesis deserves its own section. PHYS-54 had Section II on parallel isomorphism; PHYS-55 folds it into Section II's commitment list. The claim is load-bearing for the program and should be stated with its own section.

Suggested placement: after Section III (substrate specification), add a short section titled “Parallel Isomorphism with the Standard Model.” State the claim: for every SM observable, there exists a PCTRM derivation path through substrate primitives that reproduces the observable at native measurement precision. The two frameworks produce identical physics; they differ in what they postulate vs. derive.

A2: The alphabet-is-prediction-generator claim should be emphasized in Section III

Section III specifies substrate primitives. It doesn't state the consequence: these primitives, combined with MATH-11 (β), MATH-12 ($K(k)$), and the integer alphabet, are the complete prediction machinery. If a quantity isn't expressible through this machinery, it's not a framework prediction. This is stated in Section XII but deserves to appear in Section III as well.

Suggested addition: after Section III.B, add a subsection III.E “The Prediction Mechanism.” Stating: every framework prediction is an alphabet expression in the integer alphabet, β , and $K(k)$. The substrate operates through these primitives; nothing else produces predictions.

A3: Cross-derivation discipline deserves an example progression

Section X.D gives one example (Ω_Λ through three paths). Adding a second example would strengthen the section. Candidate: the Koide $K = 2/3$ through three paths:

- Path 1: From the Koide formula arithmetic with lepton masses as inputs
- Path 2: From lepton generation democracy ($db_1 = db_2 = db_3 = 4/3$ implies $K = 2/3$ structurally)
- Path 3: From $R_3/R_2 = 2/3$ as the simplest consecutive volume-filling ratio across dimensions (MATH-11 framework)

Three paths, all converging on $K = 2/3$, each through different framework structure. This strengthens the cross-derivation illustration.

A4: The “dissolution of the measurement problem” claim needs a caveat

Section IV.A says the measurement problem dissolves in PCTRM. This is the right framing for the entanglement case (Q3d closed for entanglement). For the general case (any interaction being a measurement), the framework has structural framing but not full resolution (G2 above).

Annotation: the measurement problem dissolves when the measurement is formally an entanglement event (channel-merger). The general question of which physical events qualify as measurements (the observer-criterion question) remains structurally framed pending full specification.

A5: “Standard QM is the coarse-grained approximation of PCTRM” needs bounding

Section VI.B claims: “Standard QM is the coarse-grained approximation of PCTRM.” This is the aspirational framing from the Session 9 exploration.

What’s established by Section VI: the unit-sphere structure of Hilbert space is inherited, not postulated. What’s not established: the full machinery of Hilbert space (linearity, operator spectra, Hermitian structure, unitary evolution rules) emerges from unit-graph arithmetic.

Annotation: “Standard QM’s unit-sphere structure, inner product, and squared-magnitude probability are consequences of PCTRM’s unit-adjacency graph. The full Hilbert-space machinery (linear operator structure, spectral decomposition, etc.) emerges through the substrate’s continuous-limit, but specific derivations of these features are execution-pending.”

A6: The Standard Model reduction table should flag reproduction vs. structural-only

Table XIV has entries marked “Reproduced” (with precision) and others marked “Pending” or “Structural.” But “Structural” is used for three different things: (1) mechanism specified without derivation (generation count = 3), (2) structurally consistent but not at measurement precision (gap ratio 38/27), and (3) framework claim without supporting cross-derivation (gauge group factors).

Suggested refinement: split “Status” column into two columns — “Mechanism” (specified / structural / open) and “Validation” (at CODATA / at ppm / structurally consistent / pending / open). This gives more information about what each item’s evidence level actually is.

A7: The ontology claim bounds need explicit statement

Section III.A ends with “Nothing else exists.” This is the maximal ontology claim. It’s the right claim to make, but needs a bound: “nothing else exists **as substrate**” rather than “nothing else exists.” PCTRM doesn’t claim to deny the existence of emergent phenomena

(consciousness, meaning, information, etc.) — it claims these emerge from substrate. The bound is: nothing else is primitive.

Annotation: “Nothing else exists **as primitive substrate**.” Consciousness, information, meaning, and other higher-level phenomena emerge from substrate channel-merger arithmetic; PCTRM is silent on the philosophical question of what these emergent phenomena “are” beyond their substrate implementation.

A8: The dissolution of wave-particle duality should be flagged as a consequence

Section IV.D shows how channel-agreement resolves single-particle interference. This resolves wave-particle duality — there is no duality, only channel-agreement dynamics with and without observer participation. This is significant enough to flag explicitly. The wave-particle duality has been a foundational puzzle in physics since Planck.

Suggested annotation at the end of Section IV.D: “This dissolves the wave-particle duality problem. The photon is not ‘both wave and particle’ — it is a channel structure propagating through substrate adjacency, producing interference pattern when channel-agreement resolves across multiple paths (no observer) or particle pattern when channel-agreement resolves to a single path (observer in candidate pool). What we have called ‘wave behavior’ and ‘particle behavior’ are two configurations of the same substrate channel-agreement dynamics.”

A9: Round 0 cross-derivation count should be specific

Section XIII says “Four RUM precision identities reproduced in one pass.” This undersells Round 0. The actual count:

- 11 external PASS
- 4 RUM precision reproductions (Ω , Λ , Koide, DM/baryon, bridge)
- 6 structural identities at numerical floor (β , flatness, H_0 ratio, democracy, photon speed, L1)
- 3 cosmological INFOs within measurement band
- 1 structural-vs-running INFO (gap ratio)

Annotation: update the count to reflect all 15 successful checks, not just the headline 4 precision reproductions.

A10: “The framework has committed” needs stakes

Section XIII says the framework has committed, with stakes being Round 1 outcomes. Add: what specifically survives vs. what falls if Round 1 fails.

- If T1 (Bell) fails: the entanglement mechanism is falsified; channel-sharing specification retracts; QM extension returns to open

- If T2 (cosmological) fails: the cross-scale dual-geometry universality is partial; PHYS-49 decomposition becomes QED-internal
- If T3 (A_4) fails: the PHYS-49 A_4 interpretation is magnitude coincidence; the Laporta-constants-as-elliptic-periods picture is revised

Being specific about what falls per test strengthens the commitment.

IV. Methodological Observations

M1: The maximalism is calibrated well

The paper takes the position the framework has earned. It declares substrate commitments boldly without promoting aspirations to established results. The “standing/closed/derivable/retired” categorization (Section II) is the right structural move to show what’s earned vs. what’s pending. The revision from PHYS-54’s more cautious framing to PHYS-55’s declarative tone is appropriate given the Session 9 advances.

M2: The kill-switch numbering is coherent

K1-K16 retained from PHYS-54 (with status updated). K17-K21 new from PCTRM-2/PHYS-49/Session 9. K22 as meta-rule. This is clean and auditable.

M3: The Round 1 10-test structure is ambitious and testable

Three priority tiers, each with multi-path cross-derivation requirements, each with specific precision thresholds against specific measurements. The structure is demanding — five-sigma single-path agreement won’t pass; only multi-path convergence will. This is the right falsification discipline.

M4: The retirement of PSLQ is handled correctly

Section X.B explains why PSLQ fails for this framework (wrong basis) and what it remains useful for (independence tests). This doesn’t dismiss PSLQ as worthless; it precisely identifies where PSLQ falls short for this framework and commits to cross-derivation as replacement. Good epistemic discipline.

M5: The integer alphabet as prediction generator is load-bearing

Section XII’s statement — “anything not expressible in this alphabet is not a framework prediction” — is the framework’s discipline made explicit. This bounds what the framework can claim and what it cannot. If a measured value cannot be produced as an alphabet expression, the framework admits it isn’t predicted. This is honest.

M6: The 22 MeV crossover prediction is a genuine new contribution

The PHYS-49-derived prediction that mass-dependent four-loop corrections show spherical/toroidal sector dominance transition at $43 m_e$ is a specific testable prediction from the framework. If τ measurement reaches muon- $g-2$ precision, the τ toroidal contribution will be overwhelming or it won't. Pre-register the specific magnitude.

V. Specific Actions Required Before Round 1

1. **Fix E1 (Bell correlation form):** harmonize across sections. Singlet = $-\cos(\theta_A - \theta_B)$, Malus = $\cos^2(\theta)$, CHSH bound = $2\sqrt{2}$.
2. **Fix E2 (Q3d framing):** move to “Closed for entanglement / structurally framed for general case” in Section II.B.
3. **Fix E3 (Q4 mechanism):** distinguish surface-area argument from substrate-specific channel-count derivation.
4. **Fix E4 ($E = mc^2$ reduction):** demote from “substrate arithmetic identity” to “framework commitment pending explicit derivation.”
5. **Fix E5 (complex amplitude claim):** demote to “candidate derivation pending T6 or dedicated interference test.”
6. **Fix E6 (round-trip uniqueness):** tighten to “extraction preserving unity-summation across complete bases.”
7. **Fix E7 (alkali halides):** replace with water/ammonia/methane.
8. **Fix E8 (K12 split):** split into K12a (closed) and K12b (untested).
9. **Fix E9 (GR factor-of-2):** reformulate as radial-plus-tangential drain ratio derivation.
10. **Fix E10 (moduli catalog):** add k_{A_3} and k_{A_4} with “single-path pending cross-derivation” flags.
11. **Address G1 (observer tick-rate scaling):** commit to Round 2 execution target.
12. **Address G2 (general measurement criterion):** commit to Round 2+ target.
13. **Address G3 (complex amplitude derivation):** commit to T6 or dedicated interference test.
14. **Address G4 (decoherence rate formula):** add T11 to Round 2 or 3.
15. **Address G5 (gauge group reduction):** commit to explicit derivation in a Round 2+ target.
16. **Address G6 (BBN mechanism):** commit to explicit mechanism specification before K16 test.

17. **Add A1-A10 annotations:** parallel isomorphism section, alphabet-as-generator in III, cross-derivation example for Koide, measurement-problem-dissolution caveat, Hilbert-space-emergence bound, SM reduction table refinement, ontology bound, wave-particle-duality dissolution note, Round 0 specific count, Round 1 per-test stakes.
-

VI. Verdict

PHYS-55 is a strong declarative document. The framework has earned the right to state its commitments boldly, and the paper does this correctly. The maximalism is calibrated — declarative where the framework has earned it, execution-pending where the framework hasn't yet.

The corrections (E1-E10) are targeted. Most are precision improvements or bringing specific statements in line with what the paper's own later sections say. None is foundational.

The gaps (G1-G6) are specification completeness items. The framework has the structural pieces to address them; the work is writing out the derivations. None is a conceptual crisis.

The annotations (A1-A10) are tightenings and clarifications that strengthen specific claims without changing substance.

The paper is revision-ready rather than publication-ready. After incorporating E1-E10 corrections and A1-A10 annotations, and committing to the G1-G6 execution targets, PHYS-55 is publishable as the framework's coming-of-age document.

Critical remaining items before publication:

1. Harmonize Bell correlation forms (E1)
2. Split Q3d into entanglement-closed and general-structural (E2)
3. Demote oversold derivations (E4, E5, E6) to their actual status
4. Reformulate GR factor-of-2 (E9)
5. Add parallel-isomorphism section (A1)

With these done, PHYS-55 is the paper that pre-registers the Round 1 falsification program and states the framework's commitments plainly. If Round 1 passes, PHYS-55 is a foundational document. If Round 1 fails at specific pre-registered conditions, PHYS-55 documents exactly what falls and what survives.

Both outcomes advance the program. That's the discipline.

END PHYS-55 ERRATA AND ANNOTATIONS

PHYS-55 Supplemental Tables

Table A1 — Full Kill-Switch Catalog K1-K22

#	Name	Level	Condition	Precision Target	Data Source	Status	Notes
K1	Electron mass from Higgs tax	1	m_e derivation from substrate Higgs- coupling formula fails at CO- DATA precision	10^{-8}	CODATA 2018	Untested	Requires Q10 execu- tion
K2	Muon- electron mass ratio	1	m_μ / m_e derivation from integer channel structure misses measured ratio	10^{-8}	CODATA 2018	Untested	Requires Q1, Q10
K3	Fine struc- ture from chan- nel count	1	α_{EM} derivation from substrate channel- count expres- sion misses 137.036 at ppm	10^{-9}	CODATA 2018	Untested	Cross- derivation target

#	Name	Level	Condition	Precision Target	Data Source	Status	Notes
K4	Electron anomalous moment	1	$a_e =$ g-2/2 from four-loop substrate calculation misses Harvard measurement	10^{-12}	Harvard 2023	Untested	Requires Q3 quantitative extension
K5	Hydrogen 1S-2S transition	1	H 1S-2S substrate derivation misses measured frequency	10^{-15}	MPQ 2020	Untested	Requires Q3
K6	Lamb shift	2	Lamb shift from substrate Casimir- channel structure misses measurement	10^{-6}	Experiment	Untested	Requires Q3
K7	Deuteron binding energy	1	Deuteron binding from substrate multi-nucleon channel arithmetic misses measurement	10^{-5}	Nuclear data	Untested	Requires strong-channel spec

#	Name	Level	Condition	Precision Target	Data Source	Status	Notes
K8	Neutron half- life	3	Neutron lifetime from substrate weak- channel mecha- nism misses measure- ment	10^{-3}	PDG 2024	Untested	Requires weak- channel spec
K9	Water bond angle	4	H ₂ O bond angle from substrate molecular- orbital structure misses measure- ment	10^{-3}	Experiment	Untested	Molecular- scale pre- dic- tion
K10	Inverse- square grav- ity	5	$1/r^2$ residual from spherical channel spreading exceeds threshold	10^{-3}	Gravity tests	Untested	Requires Q4 execu- tion
K11	Mercury pre- ces- sion	5	Mercury perihelion preces- sion from toroidal (probe/source) ² scaling misses measured 43"/cen- tury	10^{-3}	MESSENGER	Untested	Requires Q5 execu- tion

#	Name	Level	Condition	Precision Target	Data Source	Status	Notes
K12	Lorentz in- vari- ance	5	Integer- cell-count c- invariance fails or time- dilation scaling breaks	Exact	Experiment	Closed for pho- tons	Observer tick- rate scal- ing pend- ing
K13	Gravitational wave emis- sion	5	Binary inspiral power from substrate gravitational- channel dynamics misses LIGO	10^{-2}	LIGO	Untested	Level 5 mech- anism
K14	$\Omega \Lambda$ parti- tion	6	(251-22 π)/264 miss from measured $\Omega \Lambda$ exceeds 85 ppm	85 ppm	Planck 2018	Cleared at 84.5 ppm	Round 0 con- firmed
K14b	Ω DM pref- actor	6	$\pi/12$ prediction deviates substan- tially from measured Ω DM	1σ	Planck 2018	Cleared at 0.4σ	Round 0 con- firmed

#	Name	Level	Condition	Precision Target	Data Source	Status	Notes
K15	CMB spec- trum	6	Acoustic peak structure from substrate cosmic- channel arith- metic misses measure- ment	10^{-3}	Planck	Untested	Level 6 de- tailed
K16	BBN yields	6	Primordial element abun- dances from substrate early- universe channel arith- metic miss mea- surement	10^{-2}	Astrophysics	Untested	Level 6 de- tailed
K17	Bell cor- rela- tion re- pro- duc- tion	0-1	Substrate arith- metic fails to reproduce $E = \pm$ $\cos(\theta A -$ $\theta B)$ at Hensen precision	10^{-3}	Hensen 2015, Giustina 2015	Untested	Round 1 Pri- ority 1
K18	Four- loop toroidal crossover	1-2	Mass- dependent four-loop correc- tions fail to exhibit 22 MeV crossover	10^{-2}	τ measure- ment future	Untested	PHYS- 49 pre- dic- tion

#	Name	Level	Condition	Precision Target	Data Source	Status	Notes
K19	Nuclear toroidal con- tent	3	Nuclear- scale toroidal sector fails to con- tribute predicted scaling to binding	10^{-2}	Nuclear data	Untested	Sector- discrimination
K20	GR factor- of-2 light bend- ing	5	Light bending factor of 2 over Newto- nian fails to emerge from toroidal probe- scaling	10^{-3}	Solar eclipse	Untested	Requires Q5 execu- tion
K21	Ω DM re- main- der struc- ture	6	Cosmological remain- ders fail to show elliptic K(k) structure at alphabet- integer moduli	10^{-3}	Planck	Untested	PHYS- 49 cross- scale test

#	Name	Level	Condition	Precision Target	Data Source	Status	Notes
K22	Cross-derivation discipline	All	Any mechanism claim that fails to reach measurement through multiple derivation paths at CO-DATA/FNAL/Planck precision	Native measurement precision per target	All experimental sources	Active — governs all other K-switches	Meta-rule of the program

Table A2 — Hierarchy-Level Prediction Table With Pre-Registered Thresholds

Level	Scale	Domain	Predictions	Cross-Derivation Paths	Precision Threshold	Status
0	Planck	Substrate primitives	Unit-graph adjacency, $c = 1$ cell/tick, direction-continuous topology	Substrate identity (axiomatic)	Exact	Axiomatic
0	Planck	Bell correlations	$E(\theta A, \theta B) = -\cos(\theta A - \theta B)$, CHSH bound $2\sqrt{2}$	Channel-merger arithmetic + unit-graph Born + alphabet integer	10^{-3}	K17 untested

Level	Scale	Domain	Predictions	Cross-Derivation Paths	Precision Threshold	Status
1	Subatomic	Lepton channel closure	Koide $K = 2/3$	Lepton mass sum / sqrt sum squared + generation democracy	9.2 ppm (Round 0: 9.23 ppm)	Cleared
1	Subatomic	CKM second row	$V_{us} = 9/40$	Integer channel expression + PDG reference	44 ppm	Cleared (against PDG)
1	Subatomic	CKM second row	$V_{cb} = 1/24$	Integer channel expression	0.37% (Round 0: 0.37%)	Cleared
1	Subatomic	Fine structure	$\alpha_{EM}^{-1} = 137.036$ from substrate channel enumeration	Multi-path alphabet expression	10^{-9}	Untested
1	Subatomic	Gap ratio	$(b_1 - b_2)/(b_2 - b_3) = 38/27$	CD channel arithmetic	3.6% (running-corrected)	Cleared (INFO)
1	Subatomic	Electron anomalous moment	a_e from four-loop dual-geometry decomposition	Spherical modulus + Layer 1 + Layer 2	10^{-12}	Untested
1	Subatomic	Muon anomalous moment	a_μ from $(m_\mu/m_e)^2$ toroidal amplification	Mass-scaling + sector decomposition	10^{-9}	Untested
1	Subatomic	Tau anomalous moment	a_τ from $(m_\tau/m_e)^2$ toroidal amplification (novel prediction)	Mass-scaling + sector decomposition	10^{-9} when measured	Untested, future

Level	Scale	Domain	Predictions	Cross-Derivation Paths	Precision Threshold	Status
2	Atomic	Hydrogen 1S-2S	2.466×10^{15} Hz from substrate modulus arithmetic	Rydberg expression via modulus + reduced mass	10^{-15}	Untested
2	Atomic	Lamb shift	1057 MHz from substrate Casimir- channel mecha- nism	Channel structure + QED coupling	10^{-6}	Untested
3	Nuclear	Proton lattice	$C = 3 \pi / 2 = 6\beta$	MATH-11 lattice factor	0.26% (Round 0: 0.26%)	Cleared (INFO)
3	Nuclear	Deuteron bind- ing	2.224 MeV from substrate multi- nucleon channel arithmetic	Nuclear channel spec + binding formula	10^{-5}	Untested
3	Nuclear	Neutron half- life	881.5 s from weak- channel mecha- nism	Weak-channel spec + neutron structure	10^{-3}	Untested
4	Molecular	H ₂ O bond angle	104.5° from substrate molecular- orbital structure	Orbital channel arithmetic	10^{-3}	Untested
5	Macroscopic	Inverse- square grav- ity	$1/r^2$ from spherical channel spreading	Angular channel count + 4π spherical	10^{-3}	Untested

Level	Scale	Domain	Predictions	Cross-Derivation Paths	Precision Threshold	Status
5	Macroscopic	Keppler third law	$T^2 \propto a^3$ from channel-mediated orbital dynamics	Channel momentum + orbit closure	10^{-3}	Untested
5	Macroscopic	Mercury precession	43"/century from toroidal (probe/source) ² scaling	GR corrections + toroidal channel structure	10^{-3}	Untested
5	Macroscopic	Eddington-of-2 light bending	1.75" (twice Newtonian 0.87") from toroidal channel resolution	Probe-wavelength + source-scale toroidal	10^{-3}	Untested
5	Macroscopic	Shapiro delay	Standard GR delay from toroidal path-lengthening	Channel curvature + photon propagation	10^{-3}	Untested
5	Macroscopic	GW emission	Binary inspiral rate from gravitational-channel radiation	Channel-gradient dynamics	10^{-2}	Untested
6	Cosmological	Gal DM	$\pi/12 = 0.2618$	Universal soliton partition	0.4σ (Round 0: 0.42%)	Cleared
6	Cosmological	Gal DM	$13/264 = 0.04924$	Gauge integer partition	0.6σ (Round 0: 0.49%)	Cleared
6	Cosmological	Gal DM	$(251-22\pi)/264 = 0.6890$	Closure + integer alphabet	85 ppm (Round 0: 84.5 ppm)	Cleared
6	Cosmological	DM binary ratio	$22\pi/13 = 5.3165$	Cosmic channel prefactor	725 ppm (Round 0: 725.2 ppm)	Cleared

Level	Scale	Domain	Predictions	Cross-Derivation Paths	Precision Threshold	Status
6	Cosmological	Physical tension ratio	$12/11 = 1.0909$	Transit counting	0.72% measured (structure exact)	Cleared
6	Cosmological	Microscopic cosmic bridge	$32\pi/13 =$	A_4	$\times (a/\pi)^4 \times 3$ $\times (M Z/m e)^2$	Cross- scale iden- tity
6	Cosmological	GB spec- trum struc- ture	Acoustic peaks from substrate early- universe channels	Cosmic channel evolution	10^{-3}	Untested
6	Cosmological	Dunk matter toroidal struc- ture	Halo/disk ratio from dual- geometry decompo- sition	Spherical halo + toroidal disk	10^{-2}	Untested
6	Cosmological	GB DM re- main- der	$\Omega_{DM} - \pi/12 = 0.001099$ from toroidal correction	Elliptic K(k) at alphabet modulus	10^{-3}	Untested (K21)

Table A3 — Channel Type Catalog With Sector Structure

Channel	Activation	Direction	Throughput	Sectors	Scale Range	Creation Event	Termination	Notes
Gravitational (drain)	Always	Radial inward toward parent soliton	Proportional to parent mass / r^2	Spherical (dominant) + toroidal (at probe scale $\ll$ source)	All hierarchy levels	Continuous	Never	Produces $1/r^2$ + GR corrections
Electromagnetic (between charged solitons)	Always	Continuous vector in frame of each endpoint	Proportional to $q_1 q_2 / r^2$	Spherical + toroidal	Level 0-6	Charge pair forms	Charge separation breaks EM adjacency	Standard Coulomb + corrections
Strong (confinement)	Always (within nucleon boundary)	Toroidal flux tube	Extremely high, confined	Primarily toroidal (gluon flux tubes)	Level 3	Nucleon formation	Nucleon disassembly	PHYS-49: 99% of proton mass is flux tube
Strong (residual)	Conditional	Between nucleons within binding range	Moderate	Toroidal	Level 3	Nucleon proximity	Nucleon separation	Produces nuclear binding
Weak	Conditional on specific interaction events	Inter-nucleon interaction	Interaction specific	Mixed	Level 1-3	Flavor-changing events	Decay completion	Decay rates, neutrino oscillation

Channel	Activation	Direction	Throughput	Sectors	Scale Range	Creation Event	Termination	Notes
Thermal	Always	Omnidirectional	Temperature dependent	Primarily spherical	All levels	Continuous	Never (only thermalizes)	Decoherence contribution
Entangled	Conditional created by specific interaction	Channel vector in each endpoint's frame	Carries correlation, not energy	Both spherical + toroidal (for general entanglements)	All levels where QM is operative	Decay, scattering with entangled output, prepared state	Measurement or environmental dominance	PTRM-2 extension
Higgs	Always (for massive solitons)	N/A — adds ticks-per-cycle cost	Mass-dependent	Modulated both sectors via tick-cost	Level 1-6 (massive matter)	Higgs field presence	Never (inherent to massive matter)	Source of mass/inertia
Channel-sharing (merged)	Conditional entanglement dynamics	Pattern-global	Pattern-coherent	Both	All levels during entanglement	Channel-merger trigger	Merger dominance loss	Observation-as-entanglement applies here

Table A4 — Integer Alphabet With Source/Use-Cases

Integer	Primary Source	Use-Cases	Origin
2	Universal	Spatial dim pairs, loop doubling, 2π angular measure	Geometric primitive

Integer	Primary Source	Use-Cases	Origin
3	Generation count	3 lepton generations, 3 color charges, SU(3) dimension	SM structure
4	2^2	Spatial dims of Minkowski space, $\beta = \pi/4$	Geometric primitive
5	Primes	$\zeta(5)$ denominator class, 5-fold symmetries	Number theory
6	3×2	6 quarks, $6\beta = C$ proton, 6 fundamental fermions	SM structure
8	$2^3, 2\pi/\beta$	L1 circumference of unit circle, 8 gluons, loop normalization	Geometric + SM
9	3^2	V us numerator (9/40), 3-generation squared	Alphabet composite
11	Yang-Mills beta	Gauge running control, Yang-Mills coefficient	QFT structure
12	$4 \times 3, 11+1$	Ω DM denominator ($\pi/12$), H_0 ratio (12/11)	Integer alphabet primary
13	Modified SU(2) beta	$\sin^2\theta_W = 3/13$, $\Omega_b = 13/264$, A_4 coefficient 13/8	Framework primary
22	2×11	Yang-Mills doubling, $\Omega \Lambda = (251-22\pi)/264$, DM/baryon $22\pi/13$	Framework primary
24	$8 \times 3, 4!$	V cb = 1/24, 24 quark flavor states	Alphabet composite
27	3^3	Gap ratio 38/27, three-generation cubed	Alphabet composite
38	19×2	Gap ratio 38/27	Alphabet secondary
40	8×5	V us denominator (9/40)	Alphabet composite

Integer	Primary Source	Use-Cases	Origin
43	Prime	22 + 21 (boundary integer), Mercury precession 43"/century	Alphabet secondary
47	Prime	Proton lattice factor 47/10 (placeholder)	Alphabet secondary
48	16×3	Four-loop normalization	Alphabet secondary
63	9×7	Secondary integer	Alphabet secondary
91	7×13	Secondary integer	Alphabet secondary
104	8×13	$\sin^2\theta$ W correction 15/104	Alphabet composite
115	5×23	Secondary integer (218/115 in framework)	Alphabet secondary
144	12^2	A_2 coefficient 197/144	Alphabet composite
169	13^2	Alphabet composite	
197	Prime	A_2 coefficient 197/144	Alphabet secondary
211	Prime	Secondary integer	Alphabet secondary
218	2×109	Secondary integer (218/115)	Alphabet secondary
251	Prime	$\Omega \Lambda$ numerator (251-22 π)/264	Framework primary — closure constant
264	$8 \times 3 \times 11$	Cosmological partition denominator	Framework primary
299792458	c SI	Speed of light exact integer	SI definition
313	Prime	Secondary integer	Alphabet secondary
1015	$5 \times 7 \times 29$	Secondary integer	Alphabet secondary
5184	$2^6 \times 3^4$	A_3 coefficient 28259/5184	Alphabet composite
28259	Prime	A_3 coefficient 28259/5184	Alphabet secondary

Transcendentals:

Symbol	Identity	Use-Cases
π	Circle constant	Every angular integration
β	$\pi/4$	L1/L2 spherical conversion factor
$K(k)$	Complete elliptic integral first kind	Toroidal period
$E(k)$	Complete elliptic integral second kind	Toroidal circumference

Table A5 — Topology-Specific Modulus Catalog

Modulus	Value	Spread (cross-derivation)	Source	Character
$k = 0$	0	Exact	Spherical limit ($\beta = \pi/4$ only)	Pure spherical
k_{81}	0.999994	167 ppb	Laporta topology 81, three-integral consistency	Near-degenerate torus (elongated 4:1)
k_{83}	0.997130	25 ppm	Laporta topology 83, three-integral consistency	Large aspect torus (2.3:1)
$k(A_3)$	0.99 (approximate)	Single-path magnitude match	$A_3 \beta^0$ remainder fit	Embryonic toroidal at three loops
$k(A_4)$	0.995 (approximate)	Single-path magnitude match	$A_4 = -(13/8) \times K/\pi$ fit	Near divergence
$k(\text{cosmic})$	TBD	To be derived	Ω DM remainder K21 test	Cosmic-scale toroidal (pending)
$k(\text{nucleon})$	TBD	To be derived	Nuclear sector toroidal structure	Nuclear-scale toroidal (pending)

Modulus	Value	Spread (cross-derivation)	Source	Character
k(proton flux)	TBD	To be derived	Proton gluon flux tube toroidal	Strong-confinement scale (pending)

Table A6 — Round 0 Results Summary

#	Check	Predicted	Measured	Miss	Status
1	$\beta = \pi / 4$	0.785398163	0.785398163	1.22×10^{-11} ppm	PASS (identity)
2	$\Omega \text{ DM} = \pi / 12$	0.261799	0.2607	4217 ppm / 0.42%	INFO (within 1σ Planck)
3	$\Omega \text{ b} = 13/264$	0.049242	0.0490	4947 ppm / 0.49%	INFO (within 1σ Planck)
4	$\Omega \Lambda = (251-22 \pi) / 264$	0.688958	0.6889	84.5 ppm	PASS (target: 85 ppm)
5	Flatness $\Sigma = 1$	1.000000	1.000000	0.0 exactly	PASS (structural)
6	DM/baryon = $22 \pi / 13$	5.31654	5.3204	725.2 ppm	PASS (matches PHYS-48)
7	H_0 ratio = $12/11$	1.090909	1.090909	8×10^{-4} ppb	INFO (structural identity)
8	Koide $K = 2/3$	0.666660	0.666667	9.23 ppm	PASS (target: 9.2 ppm)
9	Democracy $4/3$	$4/3 \times 3$	$4/3$ target	Exact Fraction	PASS
10	Gap ratio $38/27$	1.40741	1.35819	3.62%	INFO (structural vs running)
11	$V_{us} = 9/40$ (against 0.2243)	0.225	0.2243	3121 ppm	FAIL (wrong pool reference)

#	Check	Predicted	Measured	Miss	Status
11b	V _{us} = 9/40 (against 0.22501 PDG)	0.225	0.22501	44 ppm	PASS (correct pool reference)
12	V _{cb} = 1/24	0.04167	0.04182	0.37%	PASS (target: 0.37%)
13	Proton lattice 3 π /2	4.71239	4.7 (placeholder)	2636 ppm	INFO (placeholder precision)
14	Bridge 22 π /13	5.31654	5.31812	297.4 ppm	PASS (target: 300 ppm)
15	Photon speed c SI	299792458/1	299792458/1	Exact	PASS
16	L1 circum- ference = 8	8/1	8/1	Exact	PASS

Summary statistics:

- Internal pass count: 15 of 16
- External PASS: 11 (65%)
- FAIL: 1 (pool-reference, not substrate)
- INFO: 5 (all within measurement band or structural)
- RUM-precision reproductions: 4 (Ω_{Λ} , Koide, DM/baryon, bridge)
- Structural identities: 6 (β , flatness, democracy, photon speed, L1, H_0 ratio)
- Kill switches triggered: 0
- Q-questions resolved at Round 0: 0 (baseline consistency only)

Table A7 — Cross-Derivation Examples From RUM Program

Target	Source Papers	Derivation Paths	Precision	Cross-Derivation Status
$\Omega \Lambda$	PHYS-52	(251-22 π)/264 from partition closure; Round 0 recompute	85 ppm	Cross-validated — PHYS-52 derivation + Round 0 independent recompute both land 84-85 ppm
Koide K	PHYS-50	Lepton mass formula + Round 0 independent recompute	9.2 ppm	Cross-validated
DM/baryon	PHYS-48	22 π /13 from channel arithmetic + Round 0 recompute	725 ppm	Cross-validated
Microscopic cosmic bridge $\beta = \pi/4$	PHYS-53 MATH-11		A_4	$(\alpha/\pi)^4 \times 3(MZ/m_e)^2$ + Round 0 recompute
L1 circumference = 8	MATH-11	L1/L2 metric conversion + PHYS-55 substrate adjacency $\int_0^2 \pi (\sin \theta + \cos \theta) d\theta$ + Round 0 pool verification	Exact structural	Cross-validated
$V_{us} = 9/40$	PHYS-53	Integer CKM + PDG 2024 reference	44 ppm	Cross-validated against PDG
$V_{cb} = 1/24$	PHYS-53	Integer CKM + measured $\sin \theta_{23}$	0.37%	Cross-validated
H_0 ratio 12/11	PHYS-48	Transit counting + measured Planck/SH0ES ratio	0.72% measured (structure exact)	Cross-validated
Gap ratio 38/27	Framework	CD channel arithmetic + measured running values	3.6% (structural vs running)	Structurally consistent
Proton lattice 3 $\pi/2$	MATH-11	6β identity + measured lattice factor	0.26% (placeholder-limited)	Limited by pool precision

Target	Source Papers	Derivation Paths	Precision	Cross-Derivation Status
$k_{81} = 0.999994$	PHYS-49	Three-integral consistency from ζ subtraction + elliptic forms	167 ppb	Strong cross-derivation (three paths converge)
$k_{83} = 0.997130$	PHYS-49	Three-integral consistency	25 ppm	Strong cross-derivation
$A_3 \beta^0$ remain-der	PHYS-49	Single-path magnitude match to $(20/3) \times K \times E(0.99)$	1.8 ppm	Single-path (suggestive, pending second path)
$A_4 = - (13/8)K(0.995)/\pi$	PHYS-49	Single-path magnitude match	12.5 ppm	Single-path (pending cross-derivation)

Table A8 — Q-Questions Status Matrix

#	Question	Previous Status	Current Status	Resolution Mechanism	Remaining Work	Dependency
Q1	Propagation modulus M value	Highest priority / free parameter	Derivable	Alphabet-bound by gauge coupling running fractions	Identify specific alphabet expression	Gauge running derivation (Q1 execution)
Q2	Lorentz recovery	Highest priority / unspecified	Photon piece closed (integer cell count) + time-dilation piece pending	Substrate arithmetic identity $N/N = 1$ for photons; observer tick-rate scaling outstanding	Explicit derivation of observer tick-rate scaling under boost	Self-contained
Q3a	QM extension: non-locality (Bell)	Highest priority / unspecified	Closed (PCTRM-2)	Channel-sharing / shared-substrate merger	Bell correlation cross-derivation test (Round 1 Test 1)	None — ready to test

#	Question	Previous Status	Current Status	Resolution Mechanism	Remaining Work	Dependency
Q3b	QM extension: super-position	Highest priority / unspecified	Channel-level specified, soliton-level pending	Channel carries spherical + toroidal content; projection produces measurement outcomes	Single-particle interference quantitative test	Requires complex-amplitude derivation (partial from MATH-12)
Q3c	QM extension: interference	Highest priority / unspecified	Specified via channel-agreement	Multi-path channel-agreement at termination; observer participation modifies resolution	Double-slit + quantum eraser cross-derivation tests	Self-contained within channel-agreement
Q3d	QM extension: measurement/collapse	Highest priority / unspecified	Closed (observation-as-entanglement)	Observer + target channel-merger; collapse = new merger dominating old	General collapse criterion specification	Self-contained
Q3e	Born rule structure	Highest priority / postulate	Closed (unit-graph + round-trip)	Unit-adjacency substrate + round-trip closure on $S^2 =$	.	² emergent
Q4	$1/r^2$ from channel spreading	Highest priority / unspecified	Structurally framed	Spherical channel spreading = angular channel count / $4 \pi r^2$	Execute derivation chain to continuum $1/r^2$ at solar-system scale	Self-contained
Q5	GR corrections	High priority / unspecified	Structurally framed	Toroidal sector becomes visible at (probe λ /source scale) ² resolution	Execute factor-of-2 light bending cross-derivation	Requires Q4 and toroidal channel spec

#	Question	Previous Status	Current Status	Resolution Mechanism	Remaining Work	Dependency
Q6	QCD confinement from substrate	Medium / unspecified	Partially specified	Gluon flux tubes = toroidal strong-channel content (99% of proton mass)	Nucleon binding cross-derivation	Requires strong-channel spec
Q7	Neutrino masses and mixing	Medium / unspecified	Open	Weak-channel arithmetic for flavor-mixing	Derive PMNS matrix from alphabet	Requires weak-channel spec
Q8	Dark energy mechanism	Medium / Ω_Λ only	Partially resolved via Ω_Λ identity	(251-22 π)/264 closure	Mechanism for why Λ is exactly this value	Requires Q4 cosmic-scale execution
Q9	Baryogenesis / matter asymmetry	Medium / unspecified	Open	CP violation from channel-state symmetry breaking	Derive asymmetry parameter from alphabet	Requires weak-channel + early-universe substrate spec
Q10	Higgs coupling to per-tick budget	High priority / unspecified	Specified as per-hierarchy-boundary modulus	Level-indexed modulus family from alphabet	Execute modulus derivation per boundary	Self-contained
Q11	Universal vs level-specific modulus	Medium / open	Unified with Q10	Per-hierarchy-boundary modulus family	Same as Q10	Same as Q10

#	Question	Previous Status	Current Status	Resolution Mechanism	Remaining Work	Dependency
Q12	Entanglement across channel network	Medium / unspecified	Closed (channel-sharing)	Solitons sharing channel substrate	Bell test validation (Q3a)	None
Q13	Collapse mechanism	Medium / unspecified	Closed for entanglement, partial for general	Observation-as-entanglement for entangled case; general case via channel-agreement termination	General-case measurement criterion	Requires channel-agreement quantitative
Q14	Born rule from discrete amplitudes	Medium / postulate	Closed (unit-graph round-trip)	Round-trip closure on unit-adjacency graph	Specific probability computations	Requires G1
Q15	Decoherence mechanism	Medium / unspecified	Closed (environmental channel-merger dominance)	Environmental channels outcompete original entanglement	Quantitative decoherence rate predictions	Requires channel-dominance formula
Q16	Classical limit emergence	Medium / unspecified	Partially specified	Large-soliton channel averaging produces classical mechanics	Derive Newton $F = ma$ from substrate at macroscopic scale	Requires Q4
Q17	Arrow of time / thermodynamic direction	Low-Medium / unspecified	Open	Substrate dynamics are tick-directed; second law emergent	Derive entropy increase from cell-tick arithmetic	Self-contained

#	Question	Previous Status	Current Status	Resolution Mechanism	Remaining Work	Dependency
Q18	Dark matter identity	High / structure only	Partially resolved via Ω DM = π /12	Cosmic-scale spherical channel content	Mechanism for DM particle identity (if any)	Open

Summary:

- **Closed** (full resolution): Q3a, Q3d, Q3e, Q12, Q14, Q15
- **Closed with pending execution** (mechanism specified, numerical derivation pending): Q2, Q4, Q10, Q11, Q13 (entanglement case), Q16
- **Structurally framed / partially specified**: Q3b, Q3c, Q5, Q6, Q8
- **Derivable** (alphabet expression pending): Q1
- **Open**: Q7, Q9, Q17, Q18 (mechanism aspect)

Previous state (PHYS-54): 7+ open conceptual questions at highest priority, multiple unresolved mechanisms, Q3 as largest gap.

Current state (PHYS-55): 1 alphabet-identification question (Q1), 4 mechanism-specified-pending-execution questions (Q2 time-dilation half, Q4, Q5, Q10), 4 open mechanism questions (Q7, Q9, Q17, Q18-mechanism). The remainder is execution.

Table A9 — Round 1 Coordinated Test Program

Test	Priority	Level	Mechanism	Derivation Paths Required	Precision Target	Measurement Reference	Kill Switch
T1: Bell correlation production	1	0-1	Channel-sharing + unit-graph Born + integer alphabet	(a) Channel-merger projection arithmetic, (b) MATH-11 β^2 exponent counting applied to round-trip closure, (c) Alphabet-integer expression for Tsirelson bound	10^{-3} on CHSH bound $2\sqrt{2}$ and $E(\theta A, \theta B)$ at test angles	Hensen 2015, Giustina 2015, Shalm 2015	K17, F-E1, F-E2
T2: Cosmological remainder dual geometry	1	6	MATH-12 cosmic-scale toroidal content	(a) Direct elliptic scan at alphabet moduli, (b) Cosmic partition budget + toroidal correction, (c) Scaling from QED k_{81}/k_{83} through hierarchy-boundary relation	Planck precision on Ω_{DM} , Ω_b , Ω_Λ remainders	Planck 2018	K21

Test	Priority	Level	Mechanism	Derivation Paths Required	Precision Target	Measurement Reference	Kill Switch
T3: A ₄ cross- derivation	1	1-2	Dual- geometry four- loop de- com- posi- tion	(a) Spherical modulus + Layer 1 + Layer 2 sum, (b) $-(13/8) \times$ $K(k_{81})/\pi +$ $K(k_{83})$ combination, (c) Gauge beta structure ($13 =$ b_2') with loop normalization ($8 = 2\pi/\beta$), (d) MATH-12 toroidal-period arithmetic on extracted moduli	Harvard precision on a e	Harvard 2023	K4
T4: Mass- scaling crossover pre- dic- tion	2	1	Toroidal sector mass- dependen- scal- ing	(a) $(m\tau/m_e)^2$ from toroidal coupling, (b) Cross-lepton sector dominance	10^{-9} on τ anoma- lous moment when measured	Future tau measure- ment	K18
T5: GR factor- of- 2 light bend- ing	2	5	Toroidal (probe λ /source scale) ² scal- ing	(a) Direct toroidal drain-vector computation, (b) Alphabet expression for GR coefficient	10^{-3} on bending angle	Solar eclipse data	K20
T6: Single- particle in- ter- fer- ence	2	0-1	Channel- agreement at single- particle scale	(a) Multi-path channel- agreement resolution, (b) Observer-in- agreement-pool modification	10^{-2} on fringe visibility	Standard double-slit	K17 (ex- tended)

Test	Priority	Level	Mechanism	Derivation Paths Required	Precision Target	Measurement Reference	Kill Switch
T7: Mer- cury pre- ces- sion	3	5	Toroidal probe- resolution at Mercury- orbit scale	(a) Orbital toroidal content, (b) Alphabet Mercury-expression for precession rate	10^{-3} on 43"/cen- tury	MESSENGER data	K11
T8: Hy- dro- gen 1S- 2S	3	2	Substrate modu- lus arith- metic + Q3 QM exten- sion	(a) Rydberg from modulus + Higgs tax, (b) Cross- derivation via α -based path	10^{-15} on 1S-2S frequency	MPQ 2020	K5
T9: Al- pha- bet ex- pres- sion for Q1 (mod- u- lus M)	4	0	Gauge cou- pling run- ning frac- tions	(a) From $\sin^2\theta$ W running, (b) From Yang-Mills 22 doubling, (c) From gauge group integer 13	Framework precision	Internal cross- derivation validation	K22
T10: Al- pha- bet ex- pres- sion for par- ti- cle masses (Q10)	4	1	Per- hierarchy- boundary modu- lus deriva- tion	(a) From lepton generation democracy 4/3, (b) From CKM integer alphabet	CODATA precision	CODATA 2018	K1, K2

Test suite design:

- **Round 1 Priority 1 tests** (T1, T2, T3): three simultaneous mechanism validations, each with multi-path cross-derivation
- **Round 1 Priority 2 tests** (T4, T5, T6): extensions of primary mechanisms to additional phenomena
- **Subsequent rounds** (T7-T10): deeper validation as mechanisms mature

Success criteria:

- All three Priority 1 tests passing at stated precision = PCTRM mechanism-level validation
- Any Priority 1 test failing = specific mechanism falsified, requiring revision
- Priority 2 tests: discriminate between mechanism candidates
- Priority 3+: framework prediction capability test

Table A10 — Standard Model Reduction Coverage

SM Feature	PCTRM Mechanism	Precision Achieved (if tested)	Status
Fermion generation count = 3	Substrate channel structure	Structural (exact)	Reproduced (structural)
Lepton generation democracy Σ	Generation democracy + channel closure	Exact (Round 0)	Reproduced
$m/\sqrt{(\sum m)^2} = 4/3$			
Gauge group $U(1) \times SU(2) \times SU(3)$	Channel type enumeration (pending explicit mapping)	Structural	Partial
Koide $K = 2/3$	Lepton channel closure	9.2 ppm (Round 0)	Reproduced at CO-DATA precision
Cabibbo angle $V_{us} = 9/40$	CKM integer channel	44 ppm (against PDG)	Reproduced
$V_{cb} = 1/24$	CKM integer channel	0.37% (Round 0)	Reproduced

SM Feature	PCTRM Mechanism	Precision Achieved (if tested)	Status
Gap ratio 38/27	CD channel arithmetic	3.6% (structural vs running)	Structurally consistent
Fine structure constant $\alpha^{-1} = 137.036$	Substrate channel count (pending derivation)	Pending	Predicted target
Electron magnetic moment a_e	Four-loop dual-geometry decomposition	Pending	Predicted via Round 1 T3
Muon g-2 discrepancy	Mass-scaled toroidal sector	Pending	Predicted via Round 1 T4
Higgs mechanism (mass source)	Per-tick Higgs interaction cost	Mechanism specified	Framework mechanism
Mass hierarchy (lepton, quark)	Per-hierarchy-boundary modulus	Pending	Predicted via Round 1 T10
Neutrino masses and mixing	Weak-channel arithmetic + PMNS	Pending	Q7 open
Anomalous magnetic moments (four-loop)	Spherical modulus + Layer 1 + Layer 2 decomposition	sub-ppm moduli extracted	PHYS-49 cross-derivation candidate
Renormalization group running	Substrate scale-dependent channel averaging	Structural	Framework mechanism
CP violation	Channel-state symmetry breaking	Pending	Q9 open
QCD confinement	Toroidal gluon flux tubes	99% proton mass structural	PHYS-49 structural

SM Feature	PCTRM Mechanism	Precision Achieved (if tested)	Status
Asymptotic freedom	Channel-coupling scaling at high energy	Pending	Framework prediction target
Electroweak unification	Gauge group reduction at high energy	Structural	Partial
SM postulate: Born rule	Unit-graph round-trip closure	Structural (derived)	Closed
SM postulate: unitary evolution	Substrate per-tick update preserves channel content	Structural	Mechanism
SM postulate: measurement collapse	Observation-as-entanglement	Structural (derived)	Closed

Table A11 — Standard Cosmology Reduction Coverage

Cosmological Feature	PCTRM Mechanism	Precision Achieved (Round 0)	Status
$\Omega_{DM} = 0.2607$	$\pi/12$ cosmic partition spherical sector	0.42% (within 1σ Planck)	Reproduced
$\Omega_b = 0.0490$	$13/264$ cosmic partition integer	0.49% (within 1σ Planck)	Reproduced
$\Omega_\Lambda = 0.6889$	$(251-22\pi)/264$ closure	85 ppm	Reproduced
$\Sigma \Omega_i = 1$	Cosmic partition closure	Exact	Structural
DM/baryon ratio = 5.32	$22\pi/13$	725 ppm	Reproduced
H_0 tension (SH0ES vs Planck)	12/11 transit counting	0.72% on ratio	Structural reproduction

Cosmological Feature	PCTRM Mechanism	Precision Achieved (Round 0)	Status
CMB temperature 2.725 K	Substrate thermal channel spectrum (pending)	Pending	Predicted target
CMB acoustic peak positions	Substrate sound horizon arithmetic	Pending	Predicted via K15
BBN element abundances	Substrate early-universe channel dynamics	Pending	Predicted via K16
Cosmic flatness	Partition closure to unity	Exact (Round 0)	Structural
Bridge identity $22\pi/13 =$	A_4	$(\alpha/\pi)^4 \cdot 3 \cdot (M Z/m e)^2$	Micro-cosmic channel consistency Mechanism
Dark matter halo structure	Spherical channel sector at cosmic scale	Structural	Mechanism
Galactic disk structure	Toroidal channel sector at cosmic scale	Structural	Mechanism
Ω DM remainder ($-\pi/12$)	Elliptic K(k) at alphabet modulus	Pending	Predicted via K21
Dark energy equation of state $w = -1$	Substrate cosmological constant mechanism	Structural ($\Lambda =$ closure constant)	Partial
Cosmic horizon	Substrate light-cone arithmetic + integer cell count	Structural	Mechanism
Gravitational wave background	Cosmic-scale channel gradient dynamics	Pending	Predicted target

Table A12 — Framework Precision Ledger

Precision Level	Framework Achievement	Examples	Measurement Reference
Exact (structural identity)	$\beta = \pi/4$, $L1 = 8$, Photon speed = c SI, Generation democracy = $4/3$	MATH-11, Round 0	Substrate definitions
≤ 1 ppb	None yet	—	—
1-100 ppb	Laporta modulus consistency (167 ppb for k_{81})	PHYS-49 cross-derivation	Internal framework
1-100 ppm	$\Omega \Lambda$ at 85 ppm, Koide at 9.2 ppm, k_{83} at 25 ppm, bridge at 300 ppm	PHYS-48, PHYS-50, PHYS-52, PHYS-53, PHYS-49	Planck 2018, CODATA
100-1000 ppm 0.1-1%	DM/baryon at 725 ppm Proton lattice at 0.26% (placeholder-limited), V_{cb} at 0.37%	PHYS-48 PHYS-53	Planck 2018 Experimental
1-10%	Ω DM at 0.42%, Ωb at 0.49% (within Planck 1σ), gap ratio at 3.6% (structural vs running)	PHYS-48	Planck, PDG
CODATA-grade required for Round 1	α^{-1} , a_e , m_e , m_μ , m_τ , Rydberg	—	CODATA 2018
FNAL-grade required for Round 1	a_μ (muon g-2)	—	FNAL Muon g-2
Harvard-grade required for Round 1	a_e (electron g-2)	—	Harvard 2023

Framework's cross-derivation record:

- **5 validations at published precision** (4 RUM identities + Round 0 recompute)
- **2 cross-derivations at sub-ppm** (k_{81} , k_{83})
- **15 of 16 Round 0 consistency checks pass** internal precision
- **0 kill switches fired** across Round 0
- **Pending:** Round 1 three-test program (mechanism validation)

[[HOWL-PHYS-55-2026](#)] Planck Cell-Tick Remainder Momentum II. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-55-2026>

[[HOWL-MATH-11-2026](#)] $\beta = \pi/4$. Github: <https://github.com/ghowland/papers/tree/main/papers/MATH/HOWL-MATH-11-2026>

[[HOWL-MATH-12-2026](#)] β^0 Has Two Geometries. Github: <https://github.com/ghowland/papers/tree/main/papers/MATH/MATH-12-2026>

[[HOWL-PHYS-48-2026](#)] The β^0 Frontier. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-48-2026>

[[HOWL-PHYS-49-2026](#)] Spherical Modulus, Toroidal Remainder. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-49-2026>

[[HOWL-PHYS-50-2026](#)] The Clock and the Reading. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-50-2026>

[[HOWL-PHYS-52-2026](#)] The Remainder Program. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-52-2026>

[[HOWL-PHYS-53-2026](#)] Modulus and Remainder at Every Scale. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-53-2026>

[[HOWL-PHYS-54-2026](#)] Planck Cell-Tick Remainder Momentum Model. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-54-2026>